Università degli Studi di Roma, "La Sapienza"

DIPARTIMENTO DI MEDICINA E PSICOLOGIA

Corso di Laurea Magistrale
in
COGNITIVE NEUROSCIENCE

BEHAVIORAL INVESTIGATION OF HEURISTIC BEHAVIOR IN A TWO-LEVEL DECISION-MAKING TASK IN MACAQUE MONKEYS

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Anno Accademico 2024/2025

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1. ACTION VALUE ASSIGNMENT AND SPONTANEOUS ACTION INHIBITION FAVORING THE UPCOMING REWARD

1.1 Rule Learning

Over millions of years of evolution, rule learning has been crucial for avoiding threats, preserving the species, and maximizing rewards (1). Applying rules was, and still is, necessary to achieve a satisfactory reward without an exorbitant use of energy. The term "rule" comes from the Latin *regere*, meaning "to guide straight," suggesting the execution of a fixed and orderly action (or thought) that can reduce the executive and cognitive cost. For example, one might always search for food in the same fruitful location, or employ the same strategy to hunt an animal. If these rules are positively reinforced, they are repeated over time; however, if the food source or hunting technique no longer proves effective (negative reinforcement), a different solution to the problem is sought. In this sense, the rule, as defined, is static and not flexible, it is a concrete directive applicable to specific situations.

To overcome this limitation, humans, as well as primates (2, 3, 4), adopt and use abstract rules. An abstract rule allows one to draw on a general principle applicable to diverse situations. Adopting this principle means establishing a causal relationship between an action, event, or target and a specific value. For example, consider the principle of mutual respect in a social group: interrupting a person's speech or action creates confusion and irritation that, in the long run, reduces interactions. In this case, the action (speaking over someone) carries a negative value. In any setting where we imagine this action, it is clear that it is applicable to every aspect of daily life. Conversely, if we are at school or university, we are expected to raise our hand to ask a question—a rule applied in specific contexts but not among friends or family. Thus, low or high degrees of generalizability define, respectively, concrete or abstract rules.

Following abstract rules necessitates adopting strategies that reduce energetic costs. One example is the so-called "repeat-stay change-shift" mechanism, in which a subject either remains faithful to a previous choice or changes it, depending on how the context evolves. This strategy has been widely described in tasks (5, 6, 7) where primates learned the stimulus-reward contingency and changed their choice of stimulus when it was no longer rewarded. In primates, however, the shift often does not occur at the

moment of reward absence. This phenomenon can be explained by Pavlov's classical conditioning; the more accurately the stimulus predicts reinforcement, the greater its inherited value. Although this process is also observed in humans, they tend to be more flexible and learn the rules of the task more quickly (8). This difference in performance in rule learning, or in the implementation of strategies aimed at learning a rule, seems to be linked to three main observations: after a rule change, monkeys tend to fixate on the previous rule, occasionally make random choices even after learning the first rule, and are less sensitive to negative reinforcement (8). The most immediate cause could be a deficit in the prefrontal cortex. Indeed, since the PFC was found to be the hub for encoding abstract rules (9), a potential deficit may lead to issues in implementing the most optimal strategy based on abstract rule evaluation. Eiselt and Nieder, then discovered that when the neurons in the cortex did not adequately distinguish between rules, the behavior of the primate on the task becomes less precise (10).

Rule learning is also strongly connected to the motivation that drives a subject to choose one action over another. In an oculomotor paradigm combined with a reinforcement learning (RL) task, an intrinsic drive in monkeys to seek predictive signals of reward has been observed, possibly driven by the need to reduce both environmental uncertainty and cognitive costs (11). Furthermore, studies have demonstrated that reinforcing the selection of targets in an operant task and motivating future choices is possible (12). When a subject takes action, they predict a likely outcome; if the option exceeds expectations, the subject updates the estimated value of that action and selects it more frequently. If the choice falls short of expectations, a reward prediction error occurs, indicating a negative imbalance between expectation and reward.

However, other factors also contribute to rule learning, with differences evident between species. Behavioral studies have shown that although humans and baboons learn simple regularities (e.g., sequence A corresponds to B) with similar ease, only humans manage to learn complex regularities (e.g., every third B corresponds to an F; 13). The difficulty animals face may lie in information processing; multi-attribute decision-making studies have discovered that, rather than focusing on integrated values, macaques tend to directly compare individual attributes (14). In this case, instead of calculating an integrated value and then making a choice, the brain directly compares individual

attributes of the options (in addition to summing them into an integrated value), which is a slower and less functional process in complex environments.

This cognitive characteristic does not necessarily prevent the animal from making difficult and non-straightforward choices. An example is the reverse-reward contingency task (15, 16, 17, 18), an inhibition task in which the animal learns to restrain its initial desire for a larger reward. While learning, it was found that presenting abstract choices (for instance, using tokens) rather than visible quantities of food helped monkeys learn faster and perform more effectively in the task (19). Another example is sequential tasks (20, 21, 22), which, in the context of learning, recreate a realistic environment and allow researchers to study how a subject learns to navigate situations where consequences are not immediate but distributed over time. In such cases, learning is often not directly causal (i.e., A corresponds to B) but requires that the subject explores, takes risks, and experiments with different strategies. Sequential paradigms introduce intrinsic uncertainty, wherein the subject is unsure if what they are doing is the right choice. This approach enables the observation of exploration, learning, and decision-making strategies that are far more complex than those seen in tasks with clear rules and immediate feedback. In short, it is possible to investigate not only whether the subject learns the rule but also, more importantly, how they construct a mental representation of the future and use that representation to guide their actions.

The maintenance of a learned rule is therefore relative to, among other factors, the subject's intrinsic motivation, the degree of generalizability of the rule, the context in which the rule must be applied, and the ability to inhibit an initial response.

According to RL theory, subjects make choices by following strategies that are sensitive to the computational structure of the task (23). In this regard, we distinguish between model-free and model-sensitive (or model-based) strategies. The former represents nearly automatic behavior in which an action is repeated because it previously led to a reward. Conversely, if to make decisions, the subject relies not solely on past experience but also on logic and the structure of the task, we are referring to model-sensitive strategies. A recent study (23) demonstrated that combining both strategies, rather than rigidly using one exclusively, maximizes rewards. According to Drummond and Niv, in model-based RL the values of actions are computed by simulating the possible

consequences using an internal map of the world. This map is defined by the states of the environment, the probability of transitioning from one state to another, and the rewards associated with each state (24). This perspective might explain why monkeys do not always change their choice immediately after negative reinforcement. Drummond and Niv hypothesize that limitations in computational resources render the model-based system more uncertain, resulting in a reversion to model-free decision control (24). This perspective might explain why monkeys do not always change their choice immediately after negative reinforcement.

In a highly uncertain and probabilistic environment, adopting a combination of strategies can help resolve the efficiency–flexibility dichotomy. For this reason, it is thought (25, 26) that subjects form hypotheses about the probability of reward and readapt these hypotheses when they are disproven, a mechanism known as the credit assignment problem.

This dynamic equilibrium establishes and ensures an adaptive basis for decision-making that considers cognitive, environmental, and physiological state factors, thereby maximizing rewards without resorting to complex computational calculations.

1.2 Brain States and the role of Cognitive Resources

Brain states are specific configurations of activity (27). Specifically, they are patterns of neural activity characterized by groups of neurons firing together in specific temporal and frequency-locked synchrony. These states emerge when assemblies of neurons coordinate their rhythms to represent perceptual objects, motor plans, or higher-order cognitive contents.

Brain states are affected by the context (28). The context in which decisions are made proves decisive in driving the subject toward one choice or another. The external context has been particularly investigated by studies (29, 30) that focused on the amount of time an animal exploited an option based on the rewards it received. In 1961 Herrnstein, who theorized the matching law: according to it, the fraction of choices allocated to each option will, at a certain point within the task session, match the fraction of total rewards obtained from that option (31). However, it's important to note that the matching law is descriptive, and without an independent mechanism or prediction, it risks circularity,

using behavior to explain behavior. Moreover, although this model explained the task they devised, it did not take into account the animal's internal motivation, determined, among other things, by its level of satiety. Some studies have since examined this aspect (32, 33). It (30) that the areas devoted to estimating how valuable an outcome is at the moment of choice (integrating the animal's brain state and continuously updating value) are different from those that focus on computing how likely it is that choosing a certain action will actually lead to a reward, thus identifying the roles of the different areas in the PFC (30). Indeed, the study of brain states cannot be separated from examination of the animal's neural activity.

One study investigating decision-making found that not only does the outcome of a previous trial influence the next one but also that trials with fast reaction times begin from different patterns of brain activity than slow ones. Moreover, brain activity changes more quickly over time in fast trials, indicating a faster progression through the neural processes that lead to a decision (34). Specifically, they studied decision-making as an emergent factor from the interaction between "starting state" (i.e., brain state) and sensory input (e.g., by varying the proportion of green/red squares in a checkerboard they could create "easy" or "hard" conditions). In their experiment, the "starting state" was simply the collective activity profile of neurons in dorsal premotor cortex (PMd) at the moment immediately before the sensory cue (i.e., the red-green checkerboard) appeared. By showing that only the initial brain state (and not the sensory tuning or the choice representation) shifts in lockstep with that outcome, they provide correlative evidence that this initial condition tracks variables like reaction time, potentially reflecting carry-over effects of attention, arousal, or reward history.
Brain states, however, is itself influenced by the external environment. If we think of the actions we perform every day, whatever they may be, we know for certain that they are strongly influenced by timing. Time dictates the rhythm of decision-making. This theme has been addressed in various studies (35, 36) which found that if the interval between trials is short, primates prefer to choose a risky option (e.g., one that does not always yield a reward). This strategy, seemingly suboptimal, is nonetheless logical: if trials follow each other rapidly, waiting a bit longer for a jackpot is less costly.

The theories of bounded rationality and ecological rationality (37, 38) have addressed this topic. The term B*ounded Rationality*, introduced by Herbert Simon (39),

describes rational behavior that is compatible with the computational limits of humans (cognitive resources, time available to decide, information available/accessible), as an alternative to the narrative of perfect rationality assumed by the *homo economicus* model. According to bounded rationality, heuristics (i.e., simplified strategies) are mental shortcuts used to compensate for human limitations.
On the other hand, E*cological Rationality* considers not just the agent but places them in close relation with the environment, thus viewing heuristics not as mere shortcuts, but as necessary tools for navigating the context effectively.

Both these theories attempt to explain how decisions are made based on context, natural or experimental, with an evolutionary approach, although many doubts still remain. One study (40), contrasting with the energy-budget hypothesis (which suggests that in periods of hunger one tends toward risky behaviors to survive, (41), found that chimpanzees are more inclined to hunt when fruit is abundant, and in laboratory contexts are more prone to risk (42). Although this contradicts Caraco's hypothesis (41), from an evolutionary standpoint it makes sense: during seasons when chimpanzees are in better condition (i.e., when they consume more fruit), they can afford the energetic cost of hunting. Ergo, risk is taken when one has the resources to compensate for potential failures. In this case, the animals did not hunt because they were hungrier, but because they had more energy available: the brain state is influenced by the external state (i.e., seasonally favorable conditions for food intake).

Reward expectation, a function encoded by the prefrontal cortex (43, 44, 45, 46, 47), has a strong impact on the animal's motivation, which is then reflected in its choices. (48) demonstrated how the prospect of returning soon to an attractive reward modulated performance even when no reward was actually delivered. According to the authors, this occurs when broader contextual motivation persists beyond motivation tied to a single trial. Indeed, we generally refer to "post-error slowing" to describe the mechanism by which humans and non-human primates slow down their actions after making an error (49). However, the adaptiveness of this mechanism has been questioned within the scientific community (50), leading to the idea that while reaction times tend to increase after errors, this may reflect a change in how information is processed in the brain rather than a strategic or beneficial adjustment (51). In fact, the study of heuristics allows us to understand how a decision has value not in itself but in relation to the agent's state. A

decision-making study on macaques, capuchin monkeys, and humans showed that although macaques and humans did not find the most rewarding long-term strategy, the strategies they adopted diverged very little, in terms of reward, from the optimal one (52). In this way, the reward was only slightly reduced at the expense of considerable energy saving. Another important observation is that ambiguous options tend to be avoided: exploring new, potentially better strategies entails not only greater energy expenditure but, above all, the possibility of obtaining no reward at all. This mechanism, known as Ellsberg's paradox (53), does not depend on culture, language, or species, having been observed even in macaques (54). Nevertheless, a recent study (55) showed that this is not always true: risk or ambiguity aversion depends on the individual, just as in humans. Specifically, the authors used a task devoid of cognitive influences such as memory and option-comparison, demonstrating that risk-taking tendency is due both to personality traits (risk-prone/risk-averse) and to the animal's state (sated/not sated). Along similar lines, a recent study (56) adopted an approach aimed at decomposing the response time of a strategy into the contributions of its cognitive components (e.g., working memory, memory retrieval, coordination, perception, etc.). Using this technique, resource demand decomposition analysis they demonstrated that the relative efficiency of the "take-the-best" strategy (evaluating attributes in order of importance) versus "tallying" (evaluating attributes in the order presented) depends on competition among concurrent activities which were making the best decision while counting in real time one, two or three bird tones. Specifically, with more attributes or high working-memory load, tallying becomes faster than take-the-best, because it incurs lower costs in updates and retrievals.

The cost of the action or of the reasoning that leads to action requires expenditure of cognitive resources. A recent study (57) investigated how attention shifts operate between internal information and external information. The basic assumption was that if working memory and perceptual attention share the same resources, the costs of switching should be similar. Indeed, the authors found that switching attention between internal and external attentional foci incurs a time cost comparable to switching between two external tasks. In simple terms, shifting attention between what we think and what we perceive has a cost, and it is similar to switching between two different perceptual inputs, suggesting that we use the same cognitive resources for both. Thus, the cognitive effort to operationalize our decisions and decide which strategy to use requires both a high level

of resources and, as demonstrated by another research group (58), a consistent expectation of reward.

Indeed, even when all information is accessible, the decisions made are often not the most optimal, precisely because of the cost (e.g., cognitive effort to pay attention or mental fatigue in inhibiting automatic responses) of acquiring the information itself (59, 60). For this reason, people tend to focus on the first option that is "good enough" rather than explore and seek a better alternative. This mechanism is called "satisficing" (coined by Herbert Simon in Administrative Behavior in 1947).

"Satisficing" can be a highly adaptive strategy (61) when gathering information has a cost: it might be defined as the right compromise between – how much I want to know something – and how much it costs me to know it..

It is thus important to consider the limits (time, attention, mental capacity, …) that accompany us in evaluating each choice and the consequent decision-making such that, when we study how humans or animals make choices, we cannot avoid remembering that these models should not be compared to a perfect and unrealistic standard, but rather taking into account the limitations we carry with us (62). By doing so, we may focus not so much on why a particular choice was made, but on why that choice was the most suitable in that context.

2. PLANNING AND IMPLEMENTATION OF SEQUENTIAL ACTIONS

2.1 Evidence on Humans and Primates

Planning is a higher-order executive function that makes it possible to define goals and the sequential actions needed to achieve them efficiently. Planning is an indispensable element of the daily lives of humans and other primates (63), although in different ways (64). It involves the ability to "think ahead" and evaluate the consequences of possible actions, modeling a sequence of steps necessary to accomplish a task (65). As Rowe et al. summarize, planning means "modeling a sequence of actions in preparation for the execution of a particular task." This process relies on the interaction of several neural systems, including those supporting working memory, behavioral flexibility, and response suppression. Rather than being a detached executive layer, planning emerges from the dynamic coordination of sensorimotor, attentional, and valuation circuits in the brain (66).

Planning is often described as a distinct phase that precedes action, followed by feedback and adjustment. However, this traditional framework, while useful in controlled lab settings, does not fully capture how decisions unfold in real-world, dynamic environments. As some authors argue (67) argue, decision-making and action are not strictly sequential but deeply intertwined, with planning continuing during movement. For example, motor planning can adjust ongoing actions in response to changes in expected reward (68), and sometimes even anticipates these changes (69, 70). A recent study (71) supports this view, showing that participants navigating an avatar along a path adapted their movements in real time, based not only on immediate affordances but also on the structure of the entire route. This highlights how motor decisions are dynamically shaped by both planning and execution processes. In the case of the RRC task, successful performance in RRC reflects not just inhibition, but the integration of value-based decision-making with ongoing action, exactly the kind of intertwined process that challenges the standard planning–execution dichotomy.

However, one aspect to consider is that planning adapts to task complexity, and the resulting actions are not necessarily the fastest, but those that allow greater flexibility in subsequent planning (72). Furthermore, it has been found (73) that the choice can be predicted by eye and arm movements before the it is made, suggesting that planning is

not separate from action, and that using the body can be fundamental for storing decisions and lightening mental load.

Theoretical models of planning are based on attentional mechanisms and hierarchical decision-making. The Supervisory Attentional System (SAS) model of Norman and Shallice (74) postulates that well-learned daily actions are governed by automatic response schemas, whereas implementing non-routine behaviors requires higher-level attentional control (the SAS). The SAS oversees the selection and activation of relevant action schemas, inhibiting those unsuitable for new goals (75). Within this model's framework, planning is viewed as the activity of the SAS proactively constructing a sequence of alternative actions and guiding the choice among them. Some subfunctions of the SAS have been localized to different prefrontal areas: for example, the dorsal frontomedial cortex appears dedicated to "energization" (initiating/sustaining a current schema), the left lateral PFC to task-setting (establishing initial stimulus–response conditions), and the right lateral PFC to continuous monitoring of output adequacy.

Another model type is the "cascade" model: Banich (76) proposes that executive functions are implemented as a temporal series of selection processes arranged sequentially along the rostro-caudal axis of the PFC (77). In this scheme, increasingly abstract representations (in anterior prefrontal areas) inhibit or activate more concrete processes (posterior areas), orchestrating the succession of decisions needed to complete long-term goals.

Hazy, Frank, and O'Reilly (2007) presented a model in which the basal ganglia provide dopaminergic gating mechanisms for updating prefrontal working memory. In this system, there is no "homunculus" deciding the plans; rather, reinforcement learning (based on dopamine-mediated reward signals) modulates the connections between the basal ganglia and PFC so that effective strategic plans can emerge. Such models can acquire, through trial and error, strategies for working memory and control, achieving performance comparable to human behavior in memory and planning tasks (78, 79). Friedman and colleagues (80) suggest that one can consider a continuum between routine and goal-directed behaviors, mediated by successor representations that reflect a hybrid form of rapid, habitual planning.

The ultimate result of planning is the implementation of single actions or action sequences. In our daily lives, often without realizing it, we constantly execute sequential actions, such as all the steps involved in making ourselves a sandwich. Sequential actions, movements ordered in time and space to achieve a purpose, form the basis of many complex skills (from writing to object manipulation) and are a focal point for behavioral neuroscience research. Planning and executing sequential actions give us control over our environment and save the effort we would otherwise expend by restarting each action from scratch. For example, if we need to make ourselves a sandwich and open the cupboard, we will also take the bread stored there rather than first taking the bread, closing the cupboard, setting down the bread, and then returning again. We execute these sequential actions because we have already planned them.

The serial order problem described by Lashley (81) highlighted that controlling sequences is not merely the concatenation of elemental acts but requires hierarchical organization and integrated control. Conceptually, motor sequences can be seen as motor programs organized in a chunked and hierarchical structure, whereby blocks of elementary movements are grouped into functional units (82). Cognitive and psychological theories have proposed that this organization results both from memory limitations (chunking originally conceived as compression of elements) and from execution efficiency (83, 84). Hierarchical integration and anticipatory planning thus allow the transformation of intention into composite actions, enabling rapid, fluid transitions between sequential elements.

Fundamental properties of motor sequences include encoding of ordinal order (assigning each act to a position in the sequence) and timing (temporal intervals and execution rhythm). Moreover, the distinction between externally guided movements (following visual or auditory cues) and self-guided movements (generated from internal memory) affects brain involvement: self-guided tasks engage planning circuits and motor memory more deeply (85).

The sequential organization of behavior has been modeled both cognitively and computationally. An influential proposal is that of hierarchical models: actions are coded at different levels of abstraction, with higher-order modules defining chunk order and lower-order modules controlling motor details (86). For example, it is hypothesized that

a spatial system (high level) codes the sequence of goals, while a motor system (low level) handles kinematic and temporal transformations. Hierarchical neural networks, inspired by Lashley (81) and subsequent work, posit the storage of sequential plans in prefrontal cortex and their transfer to subcortical circuits for automatic execution. Other models focus on associative learning: reinforcement models consider the basal ganglia responsible for temporal-difference learning, in which time coding (credit assignment) is essential for linking rewards to action sequences (87).

In cognitive science, chunking is often discussed not only as a mnemonic compression phenomenon but also as an emergent process from the interaction between behavioral variability and reinforcement-based selection. Simulations in multi-level models have shown that by combining a voluntary (cortical) control system and an automatic (basal-ganglia-cortical) system, one can capture how some transitions between movements become reinforced over time, grouping acts into ritualized blocks. Experimental data support these predictions: during motor sequence learning, output variability (pressures, saccades) decreases while total sequence time shortens (88, 83).

In nonhuman primates, motor sequences are studied using various behavioral paradigms, typically requiring animals to perform motor or ocular movements. A common model uses sequential reaching tasks in which a monkey must touch a series of visual targets with its right hand in a predetermined order (89). These tasks can be visually guided (each target appears and disappears before the next appears) or memory-guided (the monkey learns the sequence and executes it without intermediate visual cues). One example is the “repeating task” vs. “random task” of Ohbayashi and colleagues, in which identical movements were executed under visual and memory conditions (90). Another example is Matsuzaka’s experimental design (91), which shows that, with prolonged training of up to 50 days, monkeys significantly improve the speed of automated sequence execution, following typical learning curves (rapid initial improvement followed by a plateau).

Other paradigms use sequences of ocular saccades: for instance, Fujii and Graybiel (92) presented monkeys with a series of six visual targets and required successive saccades in a set order, observing neural correlates in prefrontal cortex and striatum. Paradigms like the Serial Reaction Time (SRT) task have been adapted for

primates with modifications to improve continuity: Ohbayashi (93) employed a continuous SRT in which monkeys performed a continuous cycle of sequences to maintain fluid performance and observe learning over extended periods.

In general, paradigms differ in: (1) nature of the initiation signal (self-initiated vs. external cue), (2) reliance on short- vs. long-term memory, (3) spatial vs. motor elements (the task may require specific spatial positions or non-spatial joint configurations), and (4) temporal interruptions (continuous execution vs. pauses between movements). These variables strongly influence which brain circuits are activated but also, and above all, the animal's performance.

Learned sequences are motor skills that exhibit both behavioral and neural plasticity. In Ohbayashi's study, initial learning of a new sequence (fast learning phase) led to rapid improvements especially in the spatial/conceptual aspects; in later phases (slow learning), optimization focused more on motor and temporal details (85).

Hikosaka and colleagues proposed that in the first stage, spatial information regarding the order of targets is acquired, while in the second, motor execution parameters are refined (94). This shift is also reflected in changes in brain activity: imaging studies show a shift in activation from prefrontal and associative regions toward posterior sensorimotor areas as the sequence becomes automatic (95).

Sequence learning often shows limited generalization: learned sequences tend to be tightly tied to context (specific stimuli and movements) and transfer poorly to other conditions (e.g., a new type of sensory feedback). However, once sequence chunks are learned, animals can reassemble them in new orders, suggesting some hierarchical flexibility. Behavioral experiments in primates have shown that elements of a learned sequence can be reused in different tasks, especially if the underlying structure remains similar; generalization increases in broader neural networks or circuits that encode the abstract rule rather than specific movements (96, 97).

Basic neurophysiology teaches a simple but outdated model positing that the intention–action interface can be conceived as a stack of transformations: the highest level (prefrontal) plans the sequential goal and transition rules, the middle level (SMA, PMd) prepares and temporally orders subgroups of acts, and the lowest level (M1, spinal motoneurons) executes the gestures with muscular precision. This process involves

feedforward and feedback: the brain computes anticipations (forward models) of each act's outcome and corrects errors in real time. Efference copy mechanisms transmit a copy of intended actions to sensory and cerebellar areas, allowing online adjustment of sequential movements based on discrepancies between intention and actual outcome.

With practice, motor sequences can evolve from effortful, controlled processes to automatic routines. At the beginning of learning, planning and monitoring require cognitive effort and engage prefrontal areas, SMA, superior parietal cortex, and the cortico-striatal dopaminergic system (98). Over time, performance becomes faster and less variable, and control shifts toward sensorimotor circuits.

2.2 Heuristics

As we have seen before, heuristics have the possibility to explain behaviors that may appear suboptimal. However, the notion of "heuristic" as a "practical rule" in decision-making has evolved over time. Simon, who we have seen introducing the concept of bounded rationality, suggested that heuristics could be used in models of artificial intelligence and cognition to make effective decisions under resource constraints (99, 100). In the 1970s, Daniel Kahneman and Amos Tversky launched the systematic study of cognitive biases and heuristics: their 1974 paper (101) demonstrated how probabilistic decisions employ mental shortcuts which, while useful, can produce systematic errors. They identified heuristics such as representativeness, availability, and anchoring, and highlighted the resulting distortions (for example, how the framing of a problem influence choices). In the 1990s, Gerd Gigerenzer and colleagues introduced the "fast-and-frugal" heuristics paradigm, arguing that an adaptive "toolbox" of simple rules can yield ecologically rational decisions (102). In practice, Gigerenzer showed how straightforward rules (e.g., "take the first salient cue") often produce excellent decisions.

Today, the concept of heuristics is integrated into cognitive psychology, behavioral economics, and the neuroscience of decision-making. As Hjeij and Vilks note (103), despite the variety of definitions, "all heuristics share the trait that they do not guarantee optimal solutions". In fact, when a problem is too complex or lacks a clear method for the best answer, the use of heuristics becomes inevitable. In sum, from Simon's bounded rationality to Gigerenzer's critiques, the history of research has

progressively recognized heuristics as canonical, and often unavoidable, cognitive strategies in decision-making.

Heuristics matter because they enable us to make complex decisions quickly and with little cognitive effort. Instead of evaluating every detail, the brain can apply a simple rule to arrive at a "good enough" answer swiftly. This balance between speed and accuracy is crucial in many real-world situations. For example, the "take-the-best" heuristic (104, 105) chooses between two alternatives by focusing on the first most discriminating factor and ignoring all other cues. Gigerenzer describes how airport border agents use this rule: to decide whether to search a passenger, they select the first "strong" factor (country of origin, nationality, etc.) without weighing every possible variable.

Other well-known heuristics include the nearest-neighbor rule (106), useful in route-planning problems. For instance, in foraging tasks that require visiting multiple resource sites, nonhuman primates often follow this strategy: they always move to the nearest unvisited site, achieving surprisingly efficient routes without computing the optimal path (107). Likewise, in dynamic contexts humans use "satisficing": they choose the first alternatives that meet minimum criteria, avoiding examination of the entire option set (103).

Heuristics allow decision-makers to simplify complicated problems. Although they may introduce some errors or biases, they typically lead to sufficiently good solutions most of the time. From an evolutionary standpoint, using rapid mental shortcuts can represent the best available strategy for brains with limited time or resources. Without heuristics, we might fail to reach any decision when needed or spend an unreasonable amount of time on overly complex calculations.

Many cognitive heuristics are not exclusive to humans but are also found among our evolutionary relatives. For example, Brazilian tufted capuchins (Cebus apella) exhibit the human-like framing effect: when choosing between a "safe" and a "risky" option, the way the choice is presented (as a gain or a loss) influences their preferences. In a series of experiments, capuchins treated identical food opportunities as "losses" when framed that way, choosing the risky option more often, just as humans do in gain/loss games (108).

Further evidence comes from more complex choice tasks. In an innovative experiment, Yang and colleagues (109) trained rhesus macaques to play a simplified version of the video game Pac-Man. Even in this sequential, multi-objective task, the primates clearly used heuristic strategies: the researchers' choice model revealed that the monkeys adopted a take-the-best rule, focusing at each moment on the "dominant" move and ignoring less relevant information. In other words, rather than calculating every possible long-term sequence of moves, the macaques considered only the most important factors (e.g., eating pellets vs. avoiding ghosts) and acted accordingly. This demonstrates that macaques can form and apply heuristics when facing complex tasks.

Similarly, Mochol and colleagues recorded neural activity in the prefrontal cortex of macaques engaged in a sequential decision task (110). Although the animals were instructed to focus solely on the current stimulus, they nonetheless developed strong choice biases influenced by the history of previous trials. The study showed that these behavioral biases were represented by prefrontal neurons even before the stimulus appeared, as if the brain were applying a learned heuristic to anticipate the most likely choice. In practice, even macaques "learn to be impulsive": their brains use shortcuts based on past trial history (e.g., if a certain direction was recently successful, they tend to repeat it).

In planning complex or long-term actions, heuristics play a key role in reducing the problem's entropy. When a decision-maker must evaluate future sequences of events, exact optimal calculation can be prohibitive. For example, Korn and Bach (111) showed that in multi-step foraging tasks, humans use a combination of heuristic policies and optimal strategies, balancing speed and accuracy. In their study, participants planned five successive decisions to maximize food gathered: they relied heavily on a simple heuristic but also retained some analytical computation of the optimum. fMRI scans revealed that prefrontal areas (medial and dorsal) encoded both the heuristic and optimal policies and their respective uncertainties. In practice, the human prefrontal cortex appears to integrate both strategies, using errors between the heuristic and the optimum to calibrate decisions.

In nonhuman primates, multi-step planning tasks have yielded similar results. The hierarchical decision-sequence study by Lorteije and Zylberberg et al. (112, 113) showed that macaques (and also humans) confronting a multi-level decision tree employ mixed

processes: they first integrate sensory information for all sub-goals simultaneously (in parallel), and then compose an overall strategy. This behavior, which cannot be explained by a simple "flat competition" (i.e., a model where all possible actions compete equally without any hierarchical structure) between actions, indicates that primates use hierarchical heuristics: they focus on local choices before defining the global strategy. In other words, the brain "plans in modules," solving smaller subproblems first and then synthesizing the complete solution.

In sequential tasks, heuristics play an analogous role. Consider the Pac-Man game for monkeys: the animal must make dozens of successive moves to complete a level. Analysis on the behavior showed that each time the primate chose actions by applying rule-based strategies (e.g., "follow the pellets" or "clear the ghosts") rather than planning every possible future combination of moves (109). The complex task is thus broken into phases: in each phase the most relevant heuristic is activated. This fractal, hierarchical approach observed in the game demonstrates that primates execute long sequential tasks by assembling blocks of strategies.

Other sequential tasks studied involve spatial foraging problems: in one study, monkeys were trained to visit multiple feeding stations in sequence (114). The primates showed that they repeated efficiently learned sequences through successive trials, but at times they reverted directly to the nearest-neighbor rule, especially early in learning. In general, in multi-choice tasks it is common to observe that monkeys treat a series of choices as a concatenation of local decisions according to simplified rules. If the task is to choose between two food lanes at a time, macaques may mentally handle the immediate choice first (i.e., short-term) and then integrate the result into a longer-term strategy (115).

Even when sequential tasks involve prolonged learning, heuristics are employed. In the "patch-leaving" experiment (deciding when to abandon an unstable food source for a more stable one), some researchers hypothesized a win-stay/lose-shift heuristic: if a monkey obtained good rewards ("win") from a patch, it would stay until success ceased; otherwise ("lose"), it would switch patches (116). Although in free-exploration settings this strategy did not emerge clearly, primates may learn this heuristic in early stages of serial task learning. In practice, when the environment-reward dynamics become stable,

primates seem to adopt deterministic rules like win-stay/lose-shift but maybe just up to a point.

Incorporating heuristics into decision-making models means acknowledging an agent's limited computational power and its need to use shortcuts. This approach reproduces empirical phenomena such as cognitive biases (framing effects, loss aversion, anchoring) without resorting to arbitrary assumptions. Gigerenzer's fast-and-frugal models, for example, show that simple rules can predict the real choices of humans and animals better than complex models that ignore cognitive limitations.

Heuristics thus provide a bridge between ideal rationality and actual behavior: they recognize that reasoning errors can, in fact, be adaptive.

2.3 The integration of planning and the implementation of sequential actions in the reverse-reward contingency (RRC) task

The reverse-reward contingency (RRC) task reverses the usual relation between choice and payoff: subjects are presented with two sets of rewards and must choose the smaller one in order to receive the larger reward (e.g., if a monkey sees 2 peanuts and 5 peanuts, it must choose the 2 peanuts to receive the 5. If it picks the 5, it only gets the 2; 117, 118). This counterintuitive rule creates a strong conflict: a prepotent bias favors the larger, more desirable item, yet doing so yields a smaller payoff. RRC has therefore been used widely to probe inhibitory control and related decision processes in animals. Early studies by Boysen and Berntson (1995) showed that chimpanzees readily pointed to larger amounts (getting nothing) and failed the RRC task, illustrating the difficulty of suppressing the instinct to choose "more" (119). Subsequent reviews emphasize that RRC performance reflects a compound cognitive process: subjects must evaluate relative values (i.e., the numerical assessment) and then override their preference (behavioral inhibition) to choose the lower-value option (120). In this sense, RRC performance may depend not just on inhibitory control, but on the subject's capacity to reframe the task in terms of an abstract rule or category, such as learning that selecting the smaller set consistently yields a larger reward.

Because RRC involves a sequence of operations (e.g., choose, then receive the unchosen reward), it links to broader themes of sequential planning and decision-making.

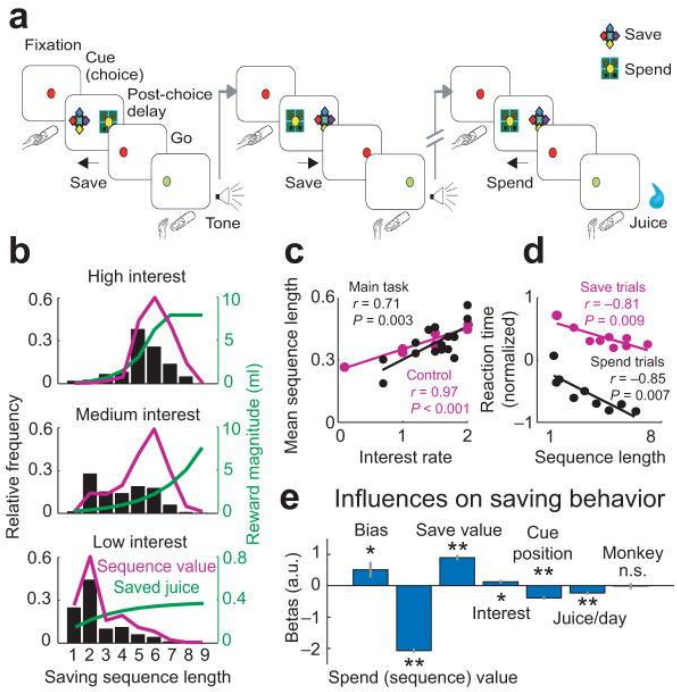

Figure 1. Monkeys can engage in internally planned sequential choices to maximize reward. In a sequential "saving" task (121). Animals freely chose "save" or "spend" on each trial, accumulating reward across a sequence. Monkeys adapted sequence length to the task's interest rate, demonstrating future-oriented planning (118).

In contrast to classic one-off choices, which are not context dependent, some primate tasks require sequences of actions or choices to obtain a delayed goal: thus, RRC can be seen as testing whether an animal can anticipate the future outcome of a single choice that reverses expectation. More explicitly sequential tasks show that monkeys can plan multi-step sequences towards distant rewards (121). In this task, monkeys repeatedly chose a "save" option to increase a hidden reward over several trials before choosing "spend" to collect it. Animals adjusted sequence length to the reward schedule, indicating internal planning (Fig. 1; 121). While this task involves gradual accumulation and future-oriented control, the RRC task tests a different challenge: whether an animal can override an immediate preference in a single moment, choosing the smaller visible reward to obtain the larger one. Thus, although both tasks involve deviations from instinctive responses, they reflect different aspects of cognitive control. The repeated failure of primates on RRC suggests that anticipating the hidden benefit of choosing the smaller option is much harder than planning in multi-step tasks: chimpanzees will delay gratification for accumulating rewards yet still impulsively pick the larger set in RRC and thus get nothing

(118). This contrast implies that RRC taps a unique decision conflict rather than just general self-control or planning.

Although RRC is generally a single-step choice, performance depends on anticipatory cognitive processes. The subject must foresee that choosing the smaller pile will yield the larger payoff. This is essentially a planning or prospective inference problem.

Beran and colleagues (118) directly compared an accumulation task (chimpazees could point to smaller items successively to accumulate larger rewards) with the RRC task: the animals easily accumulated all rewards (pointing to small sequentially) but still could not solve the RRC inversion. This suggests that planning over multiple visible trials is possible, but inferring a hidden reversal rule in a single choice is not.

Some exceptions hint at latent planning ability. In one study, cotton-top tamarins initially always chose the larger pile in a 1-vs-3 RRC, getting nothing. However, when these tamarins first experienced a training task where choosing the 1-piece yielded three food items, all subsequently learned the original RRC version (122). After that remedial experience, each tamarin solved the 1-vs-3 RRC they had failed. This indicates that the tamarins could plan or anticipate the benefit of the smaller choice once they had direct evidence of its payoff (122). In other words, prior training made the inverted contingency salient. In the absence of such experience, however, their default "plan" was just to choose large. Thus RRC performance can reflect whether the animal has explicit experience of the remote outcome.

RRC tasks create a clear decision conflict: the immediate perceptual preference (larger or more desirable food) conflicts with the rule for reward. Successfully reversing this requires inhibiting the prepotent response.

Rhesus macaques, once thought to fail RRC, can eventually learn the rule but often require dozens of trials (123). In this study, all six rhesus macaques began by selecting the larger pile but "eventually learned" to choose smaller. Their performance plateaued around chance for extended blocks, but some individuals attained criteria comparable to criterion levels in other inhibition tasks. Thus, Old-World monkeys can solve RRC if after persistent training.

2.3.1 Adding one level of abstraction

The conflict in RRC is not purely about inhibition; both perceptual salience (such as the visual appeal of larger piles) and cognitive salience, such as the learned significance or meaning of symbols, also play roles. For instance, Boysen & Berntson found chimpanzees easily inhibited to numeric symbols but not to candy piles (117). Similarly, capuchins found RRC easier when the alternative was a qualitatively different food: Anderson et al. (2008) gave capuchins a qualitative RRC (one pile of apples vs one of grapes) versus a quantitative RRC (one vs multiple same food). Strikingly, all capuchins immediately solved the quality version, whereas only one solved the numerical version without training (124). Monkeys thus may use a strategy of value dimension switching: if the choice involves different types of reward, they more readily choose the "less" (perhaps because novelty or variety is valued). In contrast, when options differ only in size of the same food, their bias is stronger. This suggests they may default "take the more familiar or larger quantity" unless another dimension makes one option stand out as better in a different way.

Another example involves causal expectations. Albiach-Serrano and Call (2014) tested chimps on a support-stick RRC: one stick could bring food, the other could not. Chimps strongly preferred the functional stick even though it violated the reward rule (125). They performed worse on this causal RRC than on an otherwise identical arbitrary task. In other words, their causal knowledge (the functional stick is better) overrode the RRC contingency. This reflects a preference bias driven by causality: chimps treat one option as intrinsically more valuable beyond just its reward contingency. Such results highlight that RRC performance is shaped by learned biases that can prevent animals from focusing on the abstract inverted payoff.

2.3.2 Adding one decision step

In sequential RRC contexts, primates often rely on simple decision rules or biases. A pervasive bias is quantity salience: even after occasional success, many animals revert to "choose more" out of habit. When allowed to iteratively choose (like accumulating tasks), a repeated strategy can form (e.g. always wait for the second item).

In tasks where repeated trials are possible, an animal might "persist" on choosing small if it just succeeded or switch after failure. However, because choosing large always

yields failure (reward goes to competitor or nothing), a naive lose–shift rule would induce them to try small after any loss. Indeed, capuchins in some studies eventually learned to pick small consistently – a strategy that could have been discovered by simple feedback sampling rather than necessarily understanding the abstract rule behind. For instance, in the token experiments, successful monkeys tended to choose the high-value token repeatedly once they discovered it paid off (19).

From a sequential-behavior perspective, it would be valuable to integrate RRC more explicitly with multi-step tasks. For example, could one embed RRC trials within an accumulation paradigm, or vice versa, to see if animals learn to switch strategies.

Lastly, RRC tasks invite exploration of heuristics in primate decision-making. Animals clearly bring biases (quantity, quality, causality) into such tasks (122, 125). Mapping these biases, for example, by systematically varying reward dimensions or context cues, would clarify how primates parse complex choice situations.

3. THE RELATIONSHIP BETWEEN ENGAGEMENT AND REWARD

3.1 Classification of Engagement in Sequential Tasks

The state of engagement in a cognitive task can be defined as the level of attention, motivation, or active involvement with which an animal approaches a task in order to achieve a goal (126). In laboratory-based, more controlled task designs, nonhuman primates show spontaneous fluctuations between periods of full engagement and periods of complete disengagement. This phenomenon reflects intrinsic variations in engagement level that do not merely depend on task difficulty or motivational load, but rather appear to represent processes distinct from those factors. Grohn and colleagues emphasize that animals periodically switch from a state of full engagement, characterized by prompt and consistent responses, to a state of disengagement, characterized by no responses or very delayed responses, even in the absence of any changes in the task or environment (126).

In the literature, the concept of engagement draws analogies to broader psychological notions such as intrinsic motivation or sustained attention (127). However, in studies on nonhuman primates, engagement is often defined behaviorally: it is the state in which the animal is actively involved in the task and oriented toward attaining the

predetermined goal and, in other words, “staying engaged is necessary to maintain goal-directed behaviors” (126). Two aspects are distinguished: true engagement, in which the animal elects to participate in the task, and performance fluctuations stemming from temporary fatigue or inattention. Thus, the decision to initiate or continue the task is viewed as a process separate from the maintenance of execution speed or accuracy. In nonhuman primates, it is hypothesized that task interruption does not simply result from the fulfillment of physiological needs, but from an intrinsic cost–benefit calculation that leads the animal to cease the task beyond mere hunger or thirst (128).

In ethology studies, engagement is measured through ecological behaviors (e.g., chosen travel routes, tool use) that reflect the primate’s adaptive mechanisms. These tasks boast high ecological validity but are more challenging to control experimentally. For example, in a foraging experiment (129), the authors reported difficulties attributing variations in engagement to isolated experimental factors, since spontaneous behaviors and environmental contexts play significant roles. Furthermore, completion rates in natural tool-use tasks also depended on external motivations such as hunger or the presence of conspecifics, beyond the experimental setting. Another limitation is that many behavioral indicators recorded in the laboratory do not translate directly to the field; likewise, measures obtained in nature, such as the number of attempts to crack a nut, depend not only on engagement but also on physical factors like strength or skill.

A further obvious limitation concerns the intervals used to define whether an animal is “engaged”: different studies apply arbitrary thresholds for disengagement. Grohn and colleagues (126), for instance, use a 3-second response latency to classify a trial as disengagement; while this threshold makes sense in their paradigm, it is difficult to generalize to other task types (i.e., more complex tasks might require longer response times even under full engagement). This suggests that no single universal indicator exists; engagement must always be interpreted in the specific experimental context.

A critical issue is distinguishing motivational disengagement from cognitive errors (i.e., a mistake in thinking, reasoning, or judgment, often caused by biases, incorrect assumptions, or misperceptions). An unmotivated animal may perform as poorly as an engaged animal confused by task difficulty. Ebitz and Hayden (130) caution against interpreting lapses as mere carelessness: they may instead reflect an intrinsic exploratory

behavior. In general, a high error rate does not always indicate poor participation but may signal alternative strategies. Therefore, engagement must be evaluated by considering the full set of indicators: for example, a high error rate combined with frequent exploratory choices might indicate that the subject is still learning and experimenting, rather than completely disinterested in the task.

The challenge remains to standardize these measures: for instance, objectively defining when an animal "disengages" would require non-arbitrary thresholds that vary by task and species. The field needs more systematic criteria to segment periods of high versus low engagement. Moreover, one must account for the animal's intrinsic motivation and its social/environmental context: low engagement in the laboratory may reflect boredom or stress, whereas in the wild, alternative opportunities could influence the measurements.

3.2 Determining engagement through Eye Data

In classic perceptual decision-making studies with rhesus monkeys, the animals fixate on a central stimulus, receive sensory information (e.g., a pattern of moving dots), and then indicate their choice by shifting their gaze toward one of the options (131). In these conventional paradigms, eye movement dynamics are often controlled and restricted (i.e., central fixation until response). However, under free or partially constrained viewing conditions, fixation behavior and saccades (i.e., shifts in fixation) emerge as an important factor in determining choice outcomes. Studies on human subjects have shown that fixations made during a decision actively influence which option will be chosen (132). These phenomena suggest mechanisms by which visual attention amplifies subjective values or the evidence of alternatives during decision accumulation. Modern decision theories often describe the decision process as a gradual accumulation of evidence leading to a decision boundary (133). In this context, visual attention is not merely a byproduct of decision-making, but can actively modulate the accumulation of information. Attentional drift-diffusion models hypothesize that a viewed option gains evidence at a higher rate than an unviewed one. Krajbich and Rangel (134) proposed that, during economic choices, fixation time amplifies the subjective values of alternatives, influencing the direction and speed of accumulation. In other words, fixating on an object

increases the evidence in favor of that object, as if gaze grants credit to the fixated alternative. This hypothesis is supported by human evidence of gaze-related choice biases: the first fixation bias (i.e., the first-seen alternative is more likely to be chosen), the cumulative fixation time bias (i.e., the alternative viewed longer overall is favored), and the final fixation bias (i.e., the option being viewed at the time of the response is selected). These findings suggest a two-stage mechanism: in the initial decision phase, gaze selectively boosts the evidence for the fixated option, while in the final phase, gaze is drawn to the already accepted option and thus reflects the imminent choice (134). Emerging literature indicates that fixation dynamics in experiments with non-human primates show mechanisms surprisingly similar to those reported in human studies. Lupkin and McGinty (132) conducted an economic task in which macaques freely moved their gaze between two objects associated with different amounts of liquid rewards. The results revealed three fixation biases in the monkeys' behavior: a tendency to choose the item viewed first, the item viewed longer overall, and the item being fixated at the time of the final decision. The findings show that primates tended to prefer the option on which they spent more total time (i.e., cumulative time bias), and in nearly all cases the final fixation coincided with the chosen option. This last finding is particularly significant: it was observed that in ~97% of trials, the final fixation (i.e., just before the saccade command) was directed at the option the animal would later select. These results suggest that gaze is not neutrally distributed but both reveals and influences the decision process in primates just as in humans. The link between fixations and decision-making is further highlighted by data on fixation duration. The same authors showed that the average duration of a fixation depends both on the intrinsic value of the fixated option and on the trial context (132). In particular, first fixations lasted longer when the fixated option had a high value (each additional juice drop increased fixation duration by about 3 ms). For intermediate fixations (neither initial nor final), duration increased as a function of relative value (i.e., the value of the fixated target minus the value of the other target). An alternative framework is proposed by Westbrook (135). During the initial decision phase (before the decision boundary), gaze reflects the accumulation of the fixated element in proportion to its value. Once a choice is internally made, a second final phase begins in which gaze is captured by the already chosen option, prolonging the final fixation.

Some studies have shed light on how informative early eye trajectories can be. Sheinberg (136) demonstrated that in an object recognition task with manual response, the animal's first saccade strongly predicted which manual choice would be made. In that task, monkeys had to recognize an image shown on a screen and indicate their response by pressing a corresponding button. Despite having no constraints on eye movements, analysis showed that the initial saccade trajectory contained information about the final manual response: the direction of the first eye movement depended on the image seen and correctly predicted the choice the animal would make. Therefore, even without being explicitly rewarded for it, macaques spontaneously developed visual exploration strategies linked to response action. Fixations and early saccades thus emerge from perceptual and decisional processes, and are not mere behavioral artifacts.

3.3 Energy-Saving in Multi-Step Decision-Making

Energy balance plays a crucial role in the survival of primates. In complex and variable environments, individuals must decide how much time and energy to invest in obtaining resources such as food or access to mates (137). This involves making choices that optimize benefits and costs: in other words, primates tend to adopt strategies that maximize net energy intake while minimizing energetic effort (e.g., moving as little as possible to reach food; 138).

Theoretical models such as Optimal Foraging Theory and the Marginal Value Theorem (139) propose that animals have evolved to maximize net energy gain. According to the Marginal Value Theorem, a primate in an optimal foraging context should leave a food source when the energy gain rate at that site drops to the average rate available in the surrounding environment (140). In other words, it will move to a new source only if the expected energy yield is higher than the current one. In practice, when rewards are equal, the site or task that requires less effort or travel time will be chosen. This optimal approach is supported by experimental observations: experiments in which animals were free to move show that they spend more time at safe food sources, that is, those with consistent rewards, rather than risky and variable ones, indicating a preference for low-risk, low-cost energy strategies (141).

From an economic perspective, behavioral choices can be described through utility maximization models. In line with economic theory, a primate evaluates each option by considering the expected reward minus the effort cost (142). Findings from this study show that macaques perform precisely this subjective subtraction: in binary choice tasks varying in reward and physical effort, the animals prefer lower-effort options when rewards are equal (142). The resulting model is one of **net utility = reward – disutility of effort**, where the latter also depends on accumulated experience (e.g., after previous effort, the subjective cost may change). The optimal energy-saving strategy, therefore, is to choose the option that maximizes this net utility. Specifically, macaques consistently rejected high-resistance options even when they provided the same amount of juice as less demanding alternatives. Capuchin monkeys involved in token exchange experiments displayed similar preferences: they maximized the total number of tokens (which translated into food) while avoiding trades that required unnecessarily high time or effort (143).

In social and cooperative contexts, energy cost takes on additional importance. Capuchin monkeys, for example, react negatively when they have invested more effort than a partner but receive a lower reward (144). Data from Brosnan and De Waal (145) show that inequity aversion increases significantly when an individual observes a conspecific receiving a better reward after exerting less effort. These findings suggest that non-human primates monitor effort in relation to outcomes: high effort anticipates a proportionate reward, and the lack of such a reward is perceived as unfair. Even in cooperation, animals implicitly evaluate their own energetic costs, modulating task distribution and resource sharing behaviors accordingly.

In free foraging, it is consistently observed that primates optimize travel paths and collection times to maximize energy yield. Chimpanzees, among others, plan low-cost routes to distant fruit trees, often taking paths with minimal incline to conserve energy (146). These same chimpanzees reached out-of-sight targets by taking less direct but more energetically efficient paths, potentially indicating the use of spatial maps and evaluation of physical costs in travel planning. Similarly, lab-based foraging experiments show that macaques stay longer in patches with stable returns and leave earlier from those where energy yield drops rapidly, as predicted by the Marginal Value Theorem (140).

A paradigmatic context for multiple decisions is “patch foraging,” where the animal must decide when to harvest accumulated resources or move to new sources. In such scenarios, the optimal foraging rate theory predicts that the animal will work until the instantaneous reward rate drops below the average gain. Consistent with this, Hage and colleagues (147) showed that marmosets adjust their work duration and movement speed based on harvesting costs: when the extraction cost increases (e.g., the food tube is farther away), the animals work for longer periods, accumulating more resources, but reduce their movement speed. In other words, they prolong collection but decrease vigor, conserving energy when marginal return declines. As travel distance increases, marmosets increase the number of trials executed during each work phase but show a decrease in the speed of reward-relevant saccades. This behavior aligns with foraging optimization theory: sacrificing part of movement speed to maintain a high overall energy return: as also indicated by Korbisch (148) in relation to saccades, motor components (vigor) appear to encode real-time assessments of movement utility; during deliberation, saccade speed reflects the increasing expected value of the chosen option.

However, it's important to note that primates are not always perfectly rational: in some experiments, deviations from optimal principles are observed. A paradigmatic case is the sunk cost effect: in a psychomotor cursor task, many capuchin and rhesus monkeys persisted unnecessarily in an already-started action, even when it was suboptimal to do so (149). In these cases, the animals continued longer if they had already invested time, contradicting the energetically optimal choice of quitting early.

Finally, a critical point to consider is the difference between laboratory and natural settings. Some findings (e.g., the strong tendency to maximize gain in traditional laboratory settings) do not exactly translate to real life. In a study on macaques, Eisenreich showed that while subjects seemed risk-neutral in the lab, they became risk-averse and behaved more like humans in a naturalistic task (141). This suggests that the experimental context can influence observed strategies: tasks that are too abstract or repetitive may obscure real ecological constraints.

3.4 Using Reaction Time to Define Engagement States

Reaction time (RT) is, in cognitive psychology, a fundamental measure of the speed of response to stimulus. Welford (150) defines it as "the interval of time between the presentation of a stimulus and the initiation of the voluntary motor response". Generally, a distinction is made between simple reaction times (i.e., one stimulus and one possible response) and choice reaction times (i.e., multiple stimuli and multiple possible responses). In multi-step tasks for primates, RT can refer to the response latency at each decision-making stage of the task (e.g., cursor movement, lever release, or eye saccade toward a target) following a visual or auditory stimulus. Theoretically, reaction time reflects the sum of processes such as stimulus perception, decision processing, and motor action initiation. In other words, variations in RT can indicate changes in task difficulty, attentional allocation, or the animal's motivational state. For example, in complex decision-making tasks (151), it is known that reaction times increase with higher memory load, task difficulty, or response distance (in line with Hick's law, which states that response time increases with the number of choice alternatives; 152).

As previously discussed, the concept of engagement in experimental tasks refers to the degree to which an animal remains attentive, motivated, and committed to performing the task. An engaged animal is actively involved and promptly responds to experimental demands; conversely, a disengaged animal may ignore stimuli, respond randomly, or not respond at all. As noted, engagement is not a stable condition but undergoes intrinsic fluctuations: during experiments, animals spontaneously and repeatedly alternate between periods of full engagement and disengagement from the task (126). This alternation is reflected in behavioral markers and temporal measures, including reaction times. Generally, it is assumed that a primate highly involved in the task will show short and consistent RTs, whereas a drop in engagement may manifest as unusually long RTs or a failure to respond within the time limit. In primate literature, engagement is not strictly defined but is inferred from performance: task engagement is indicated when the animal's actions are consistent with the task structure, and task disengagement when participation significantly declines (126).

3.5 Evidence-Based Studies

There are some common trends and differences in the use of reaction time as an indicator of engagement in multi-step tasks. A first observation is that there is no single methodological approach: in some experiments, the animal can respond whenever it wants (153), while others impose a minimum fixation period or set a fixed time interval before allowing a response. In the latter case, the RT concept is less variable because the action is conditioned by waiting times. For example, in memory tasks such as the Sternberg test adapted for primates (154), subjects must maintain steady gaze until the appearance of the target; reaction times are then measured at the final signal but may be forced into a narrow window. In such tasks, studies usually focus on response accuracy rather than reaction time itself. Conversely, in exploratory or fast decision-making tasks (as in reward retrieval paradigms by Hernádi (155)), primates are free to respond immediately, making RT a more meaningful indicator of willingness to act. These protocol differences affect how RT is interpreted: in spontaneous reaction tasks, a prolonged RT may signal disengagement, whereas in structured tasks, a long RT may simply result from a different planned waiting process.

In terms of quantitative results, RT patterns vary widely depending on the paradigm and species. For instance, Beran and Parrish (156) reported typical RTs around 1.5–2.5 seconds for stimulus selection in capuchin monkeys, with significant increases in shift trials (i.e., trials in which task rules change). Hernádi (155) found that average response times in macaques decreased with increasing sequence value (i.e., the perceived value of the reward). In both cases, inter-subject variability exists, but the main effects remain robust. There seem to be inter-species differences, with macaques responding faster than capuchins, possibly due to differences in risk tolerance or food motivation (157). However, there are no systematic studies that directly compare RT and engagement across different species using identical tasks.

In the case of humans, the presence of attentional conflict leads to slower reactions, but this does not necessarily indicate disengagement; rather, it may reflect deeper processing of conflicting information. Similarly, in primates, Hassett and Hampton (158) do not interpret the slightly slower responses of macaques to incongruent stimuli as disengagement, but as evidence of increased cognitive effort. This may suggest that the

meaning of reaction times varies depending on the experimental context: in some cases, longer RTs indicate reduced motivation, while in others, they reflect increased difficulty or confusion, not disengagement.

Reaction time allows detection of subtle vigilance fluctuations that a binary engaged/disengaged analysis would miss. In some tasks (155), the use of RT to infer motivation aligns with economic theories of reaction time as a measure of motivational response vigor (159). Nevertheless, primates do not always interpret reaction times the same way humans do: macaques are known to be cautious in some choice tasks, which results in relatively high RTs even when they are fully involved. A monkey may deliberately choose to wait longer to increase its accuracy, resulting in slow RTs that do not reflect disinterest. This complicates the interpretation of RT as a simple measure of motivational involvement.

Finally, it should be emphasized that engagement is not a homogeneous construct in itself. It may reflect either reward motivation or sustained attention. These two aspects may have opposing correlations with reaction times. For example, a satiated primate may place less value on the reward and respond more slowly or not at all, indicating low motivational engagement. Conversely, even a hungry primate confused by the task may show high motivational engagement but display elevated RTs due to cognitive uncertainty. Without independently measuring motivation and attention, reaction time alone cannot distinguish these cases. This makes it essential to interpret RT results within the specific experimental context and ideally to integrate them with other behavioral indicators such as fixation time, number of saccades, and the type of choices made by the animal.

4. EXPERIMENTAL RESEARCH

4.1 METHODS AND TASK

4.1.1 Animals

Two adult male rhesus monkeys were examined (Macaca mulatta; Monkey B, 7.5 kg; Monkey C, 8 kg). All experimental procedures, animal care, housing, and surgical procedures conformed to European Directives (86/609/ECC and 2010/63/UE) and Italian laws (D.L. 116/92 and D.L. 26/2014) on the use of nonhuman primates in scientific research and were approved by the Italian Ministry of Health. Throughout the experiments, the daily monkeys' water supply was accurately controlled and adapted to the body weight. PC monitor (LCD, refresh rate 85 Hz, 800 × 600 resolution) that was equipped with a touchscreen (Micro-Touch, sampling rate of 200 Hz) was placed in front of the monkeys to present stimuli and inspect touch positions (monitor-eye distance 21 cm). A non-commercial software package, MonkeyLogic (http://www.nimh.nih.gov), was used to regulate stimuli and behavioral responses and collect behavioral data. Eye movements were recorded using an infrared tracking system (Arrington Research, Scottsdale, USA) at a sampling rate of 220 Hz.

4.1.2 Task Design

Experiments were performed in a dim, sound-attenuated room. The monkeys were seated upright in a chair with the head fixed and performed the task using the right arm, while the other arm was restrained in a comfortable position. Monkeys performed a two-level reverse-reward contingency task (fig. 2). The trial began when the central target, a white horizontal bar located at 0, -8 pxl(horizontal axis, vertical axis) appeared in the center of the screen. The monkey had to touch and hold the central target (CT) within 2s and for a random interval between 200 and 500 ms (to avoid movement anticipation). The touch tolerance window for the CT was 6°. Upon holding the CT, two peripheral stimuli (PTs) appeared on both sides of the screen. The exact positions were left: -10,1, and right: 10,1. The appearance of the two targets was randomized between left and right. The two PTs

were associated with a fixed reward of either 1 or 3 drops of reward (water). Upon the occurrence of the two PTs, the horizontal bar disappeared triggering the go-signal. Then, the monkey had to choose one of the two options by touching either one or the other PT, and hold it for 500ms. The touch tolerance window for the PT was 9°. The peripheral holding time was followed by an inter-level interval of 500ms, after which the CT appears again. The CT should have been held again to make the other two PTs appear. The two stimuli appeared are different from the ones at the first level. Moreover, they were also different within level 2: the monkey will find two different sets of stimuli whether they chose low (1 drop of water) or high (3 drops) reward at level 1. If the animal chose low reward at level 1, at level 2 one stimulus will be associated with low reward (5 drops of water) while the other one with high reward (9 drops). On the other hand, if the choice at level 1 converged on high reward (3 drops), at level 2 one stimulus will dispense low reward (0 drops) while the other one will deliver high reward (1 drop).
Each trial was followed by a period of 2 s inter-trial interval, during which the video screen was black.

A correct trial is defined as a trial in which the monkey made a choice at both levels. Within error trials it is possible to identify trials where the animal is not holding the central target at level 1, not choosing any stimuli after the holding period, and trials where he didn't make any decision at level 2. In error trials, the reward was not delivered. After 60 correct trials the set of stimuli used was changed with new ones thus making a new block to start.

Depending on the choice at the first and second level we identified four Categories: HighLow (high reward at level 1, low reward at level 2), HighHigh (high reward at both levels), LowLow (low reward at both levels), LowHigh (low reward at level 1, high at level 2). Choosing a low reward at level 1 (1 drop) ensures greater rewards at level 2 (either 5 or 9 drops of reward). Specifically, HighLow trials have a reward rate of 10 drops, HighHigh 6 drops, LowHigh 5 drops and HighLow 3 drops.

4.1.3 Task Difference between Monkeys

While the task already described resembles the one administered to Monkey C, Monkey B underwent a slightly different one. In particular, the main difference concerned the definition of block. In this case, the block change occurred after the monkey performed 30 trials on the same category, without category switches.

In other words, in Monkey C's task, each block (i.e., set of stimuli) remained active until the animal accumulated 60 correct trials **across all categories**; any combination of correct responses would trigger a block transition once the 60-trial criterion was met. By contrast, in Monkey B's task, blocks changed after the animal produced 30 correct trials **within a single category**, requiring them to repeatedly select the same stimuli until they had reached the within-category accuracy threshold.

4.1.4 Behavioral Measures Recording

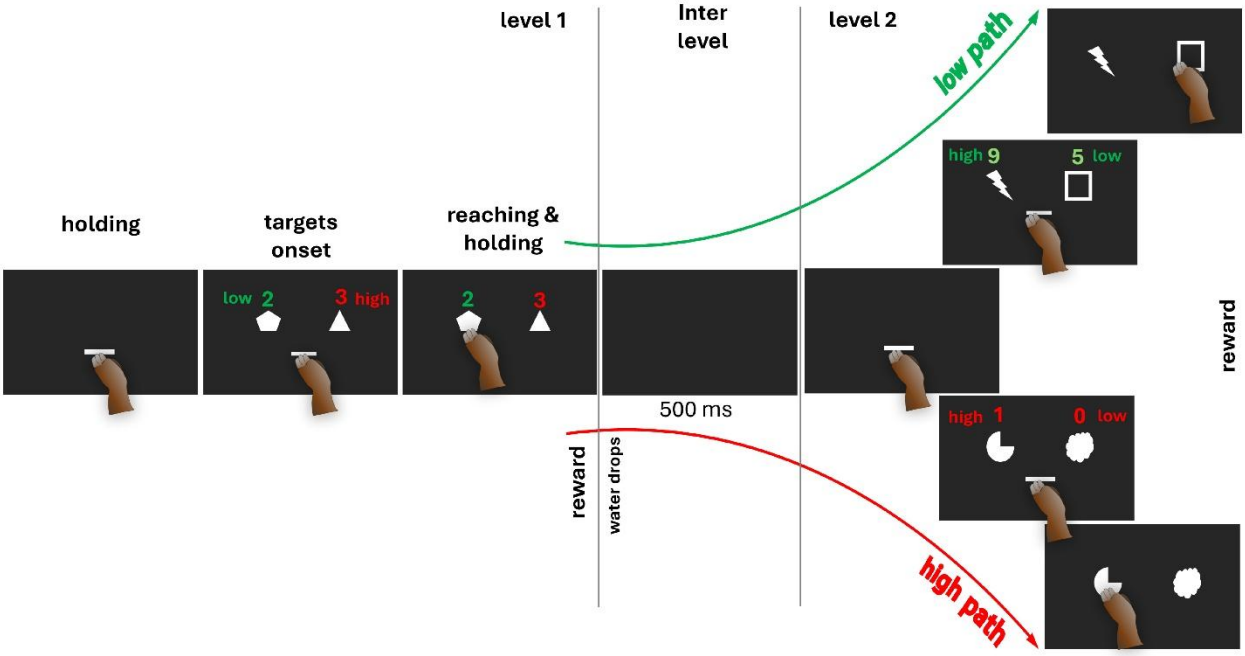

Figure 2. Reverse-reward contingency task. A central target appears for 400ms, followed by two stimuli (level 1). After the choice at level 1, reward is given. Then, two routes are possible: low path and high path, depending on the amount of reward chosen.

Four behavioral variables have been recorded during the task: reaction time, fixation time, number of saccades, and vigor. Reaction time, for each level, is the time between the CT offset and the reaching movement onset. Fixation time is, for each level, the time the gaze is detected within the fixation window from PTs onset to PT touch. The fixation window

has the same area as the touch window of the peripheral stimuli (9°). Saccades were defined using a threshold after PTs onset (speed threshold of 29 deg/sec). Vigor is the mean velocity (pxl/ms) of the reaching movement.

4.1.5 Engagement Labeling

Six categories have been identified arbitrarily based on behavioral observation: Exploration, Full Engagement (Exploitation), Level 1, Level 2, No Engagement, No Exploitation.

Trials have been labeled as Exploration when at least 3 different categories (HighLow, HighHigh, LowLow, LowHigh) were chosen within the first 10-15 trials but not if this pattern is broken by another pattern. In most cases the animal needed 5-10 trials of Exploration before sticking with a category. In some blocks, the monkey chose from the second or even the first trial, the category with which he stuck for the entire block. However, since at the beginning of the block the monkey does not know the association between the new stimulus and the reward value, we labeled the first trial of each block as Exploration.

Trials have been labeled as Full Engagement (Exploitation) if the animal chose the same category for more than 3 trials consecutively. Trials where the animal chose between 3 or 4 categories alternating among the three or four without precise patterns have been labeled as No Engagement. Trials labeled as “Level 1” are those where the animal is tracking always the same stimulus at level 1 but not the same at level 2 (with same category for max 3 trials). The opposite occurred for trials labeled as “Level 2” where the animal chooses randomly at level 1 while tracking the same stimulus at level 2. “Level 1” and “Level 2” patterns are labeled this way only if they are present for at least 5 trials alternating. Finally, error trials were labeled as No Exploitation starting from 3 errors in a row, meaning that the animal is not doing the task, or doing it without effort.

4.2 Results

4.2.1 Engagement States : Exploration, Exploitation, No Engagement

Incorrect trials were excluded from both datasets (2.76 % for Monkey C, 72.57 % for Monkey B).

To determine whether the behavioral data actually reflected different internal states, we examined the differences among trials classified as exploration, exploitation, and no-engagement at both task levels.

Because the variances differed across engagement labels (Levene's test for equality of variances: stat = 18.77, $p = 3.6 \times 10^{-5}$), we used Welch's t-test to assess whether exploration and exploitation trials differed significantly. Figure 3 shows that reaction times (both levels), fixation times (both levels), and number of saccades (first level only) all differed significantly ($p < 0.01$), whereas vigor did not. Interestingly, the mean-level trends were the same for both exploration and exploitation: reaction and fixation times increased at the second level, while saccade count and vigor decreased, with all measures being lower in exploitation trials. This finding aligns with our initial expectation that, during exploitation, the monkeys were more engaged and, knowing which choice to make, responded quicker. That speed manifests in shorter stimuli-fixation times, faster comparison between options, and more rapid decision execution.

We then asked whether trials labeled as exploration shared similarities with those labeled as no engagement in the behavioral measures. By doing this, we sought to determine whether exploration trials truly reflected an actively engaged exploratory state, or whether they instead resembled a no-engagement state. The results (Figure 3) showed that for measures such as reaction time and fixation time, there was a significant difference between exploration trials and trials classified as no engagement. Surprisingly, between exploitation trials and no engagement ones, the only significant difference emerged at the second level for reaction times. Thus, whereas behavioral measures can distinguish between exploration and exploitation phases, they fail to differentiate exploitation from no engagement. This finding suggests, despite the limitations of manual labeling, that engagement versus no engagement is not reflected in these behavioral variables.

Defining exploration and exploitation states in Monkey B's task is more complicated. Because there is no clear boundary marking a block shift, and we do not observe major changes in the animal's strategy (i.e., it tends to stick with the first strategy it finds), it is difficult to divide the trials into distinct engagement states.

Assuming the two animals' inferring levels were comparable, the changing stimuli set (which switched at each block change) evidently served as salient factors that kept Monkey C engaged. Moreover, the fact that stimuli changed every 60 correct trials, regardless of the strategy chosen, forced Monkey C to make the most rewarding choice before the block shift. That block shift itself imposed a significant cognitive effort, since the stimulus–reward contingency had to be re-inferred (relearned) each time.

In Monkey B's case, however, this restrictive decision context didn't exist. The animal could continue making decisions indefinitely, so long as it didn't stick with the same strategy for 30 trials. Given a less defined task with no sudden changes, why didn't

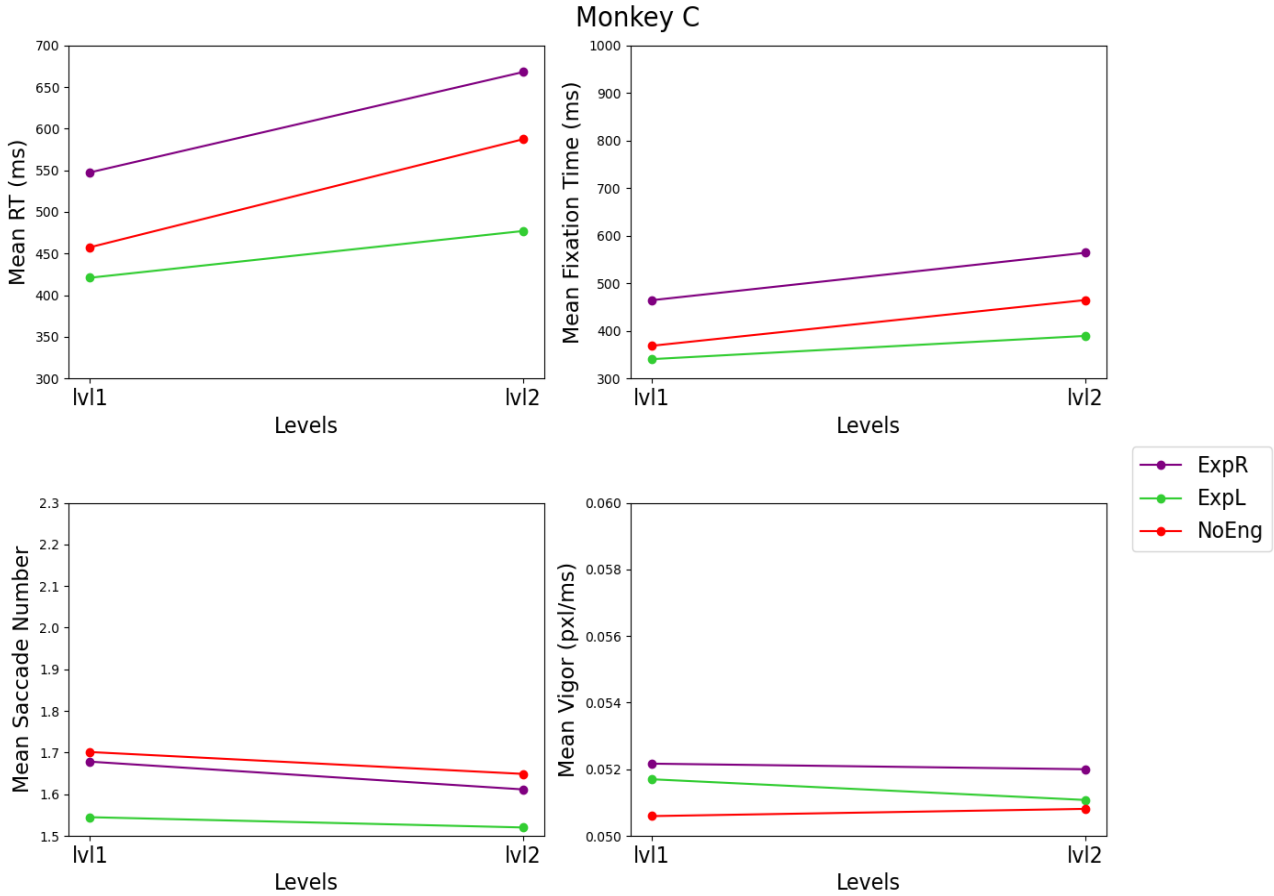

Figure 3. *Reaction Time (top left), Fixation Time (top right), Saccade Number (bottom left), and Vigor (bottom right) at both levels, all sessions. Significance difference has been found in all variables but Vigor and Saccade Number (ExpR vs No Eng, both levels).*

Monkey C and B simply identify the most rewarding category and stick with it until satiety?

Moreover, if we examine the number of correct trials per session, Monkey C averages 362.7 (SD = 5.43) while Monkey B averages 118.5 (SD = 24.6). Moreover, Monkey B shows a tendency to not start the trial, after a block change. In particular, 78% of trials in blocks after the first one are error trials (i.e., trials where the monkey is not making decisions at all) against 48.05% of the first block. This pattern has not been found in Monkey C where only 5.71% of trials were errors in blocks after the first and 2.91% in the first one. Therefore, there is a huge difference in the animals' engagement levels. Relearning the contingency is not sufficiently stimulating to overcome the decision-avoidance barrier. Considering also the negligible number of trials classified as exploration (since there is no genuine, systematic block shift to promote exploration), the only meaningful state difference to investigate is that between exploitation and no engagement, as reflected in the behavioral measures. As shown in Figure 3, the only significant difference remains in reaction time, which is slower in "no engagement" trials than in "exploitation" trials.

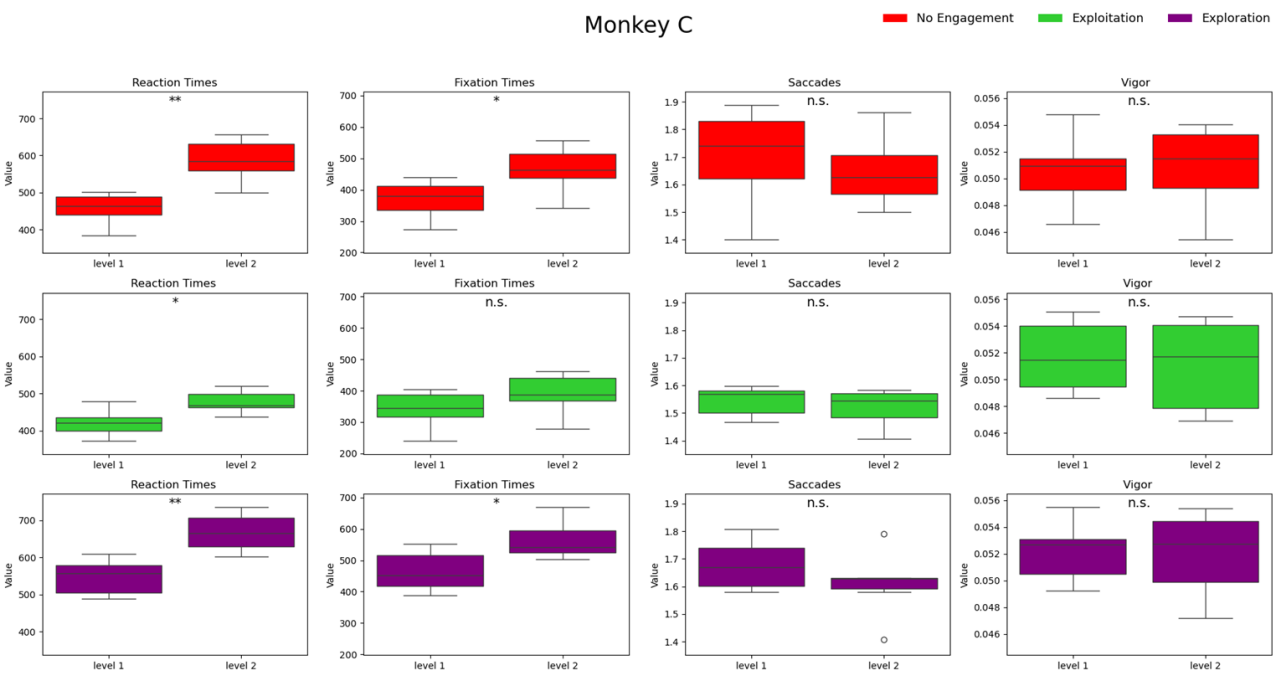

Figure 4. *Monkey C. From left to right: mean RT, Ft, saccade number, vigor for the three conditions (No Engagement, Exploitation, Exploration).*

It is interesting to note that, for Monkey C (but not Monkey B), across the two task levels there is a downward trend for variables such as number of saccades and vigor, and an upward trend for variables such as reaction time and fixation time, in all categorizations thus identifying longer values between levels. This finding is notable when we consider that at the first level (critical for the future choice, and thus the most consequential) the monkey makes decisions more quickly, evidenced by fast reaction and fixation times and by high vigor, yet performs more saccades, as if to explore the options more thoroughly. Contrary to our hypotheses, the monkey is less deliberate in its initial decision, despite supposedly needing to invest more time there. We therefore conducted a statistical analysis (Welch's t-test) to determine whether the differences between levels in the behavioral variables were statistically significant. The results (Figure 4) show that the difference is significant for reaction time in all three conditions, and for fixation time in the exploration trials and in those labeled "no engagement." With respect to saccade count and vigor, no significant differences between levels emerged. Although the difference in reaction time between levels is significant, reaction times remain, as previously noted, much lower than in other trial categories. This disparity is expected, given that when the monkey is engaged it should choose more quickly and accurately. Regarding vigor, the only significant result observed is that it tends to decrease over the course of the session, as shown in figure 5. This was found specifically at level 1 even though it was also observable at level 2.

A key difference lies in the task levels: unlike Monkey C, where there is a significant difference between Level 1 and Level 2 in reaction time (and in fixation time for "no engagement" and "exploration" trials), the same cannot be said for Monkey B, in whom neither reaction time nor fixation time showed significant differences between levels (Figure 6).

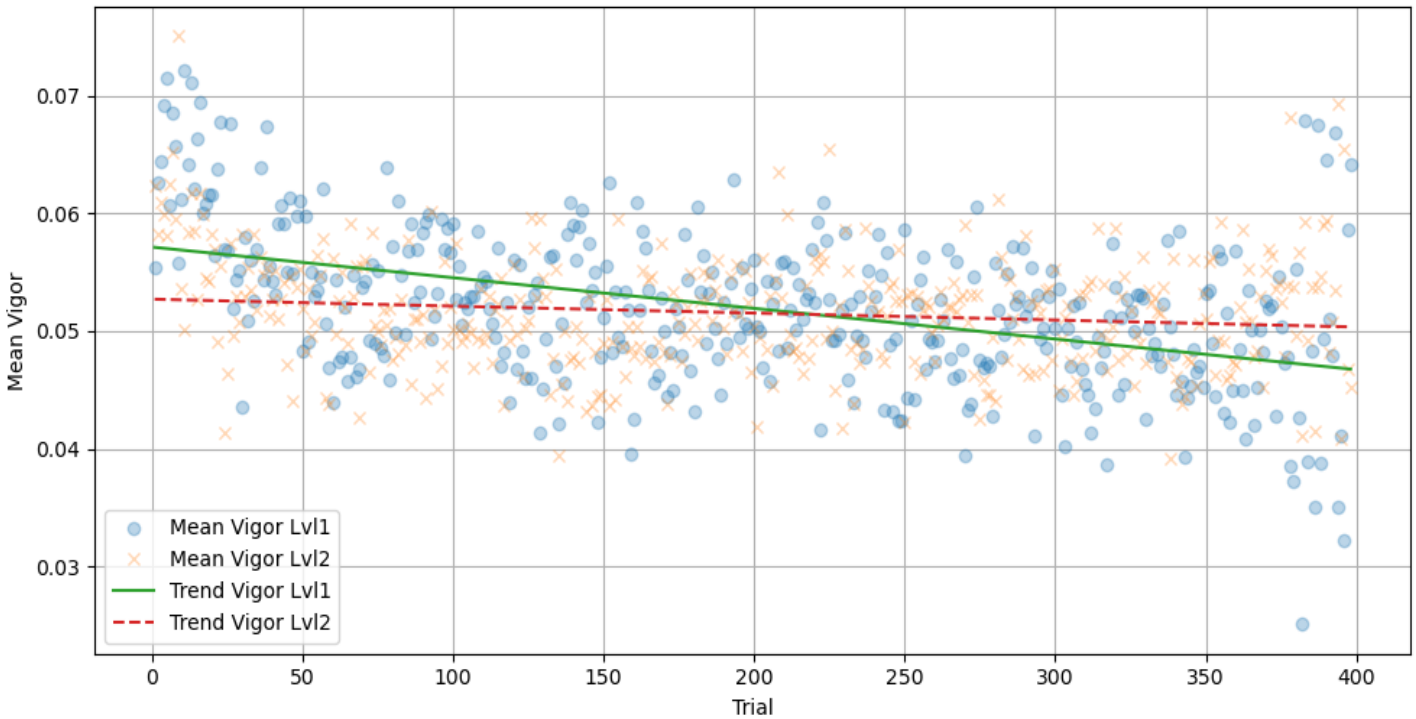

Figure 5. *Trend of Vigor at both levels over the session. A great decrease can be seen at level 1, while a slightly one at level 2.*

On the one hand, the lack of a level-dependent difference in reaction time during "exploitation" trials is understandable, the animal, already knowing which choice to make, shortens its response latency. What is harder to explain is the complete absence of level effects on the other behavioral variables. One binary hypothesis might be that the monkey occupies either a high-engagement state during "exploitation" trials or a low-engagement state during "no engagement" trials. In this view, shifting between high and low engagement could even be optimal: when engaged, the monkey minimizes reaction times and cognitive effort to maximize reward; when disengaged, it falls back on a static motor pattern of target comparison/fixation, modulating its response speed accordingly.

As Figure 6 shows, no level-dependent differences are significant for Monkey B. This suggests that, in this task, the degree of engagement does not depend on the amount of reward obtained at either the first or second level.

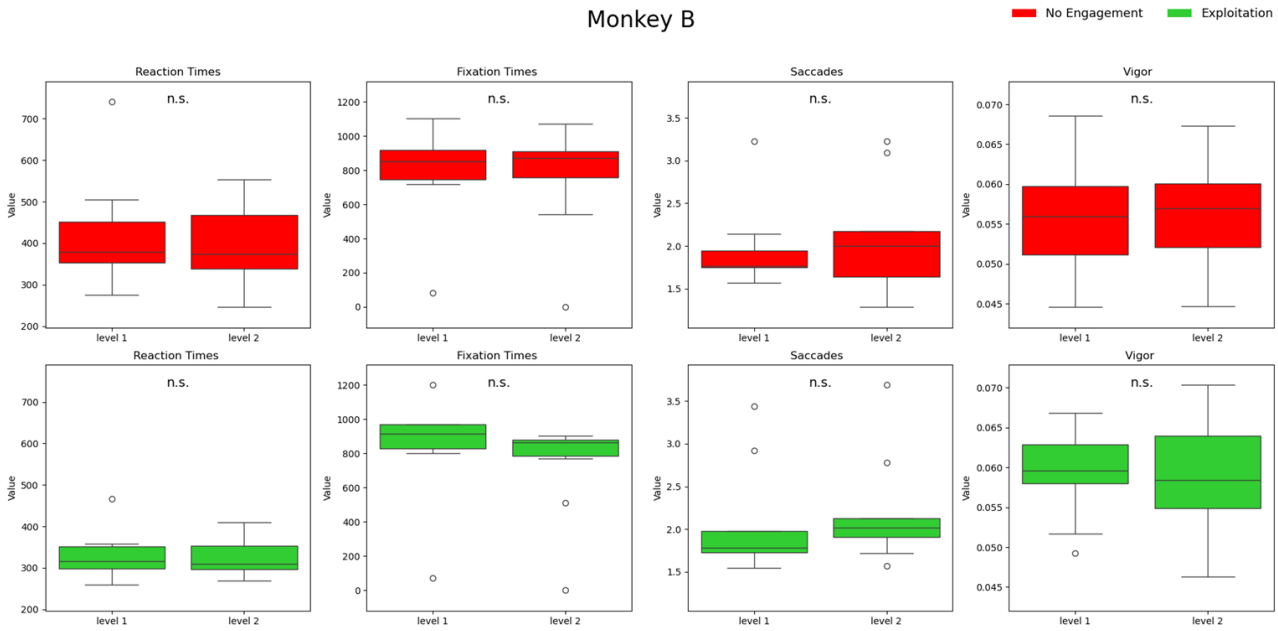

Figure 6. *Monkey B. From left to right: mean RT, Ft, saccade number, vigor for the two conditions (No Engagement, Exploitation).*

Hereafter the results concern Monkey C

Given the greater variety of behavior shown by monkey C, likely due to a more complex and structured task, the following analyses will focus on the data from monkey C.

4.2.2 Level Disengagement: being faster ensures reward but not the most optimal

To uncover the mechanism underlying why reaction times in Monkey C's task are faster at Level 1 for all trials, we asked whether there was a shift during the session such that reaction times lengthen at Level 2 because the monkey, at a certain point may become satiated and thus less engaged. We also checked whether or not the pattern was stable across sessions and if was instead driven by the course of individual trials; therefore not associated to a specific pattern.

For each group (no engagement, exploration, and exploitation), we examined the progression of reaction times (and fixation times) within each session. Figures 7, 8, and 9 show a difference between Level 1 and Level 2 with inter-session variability, but reaction times at Level 1 are more stable than those at Level 2.

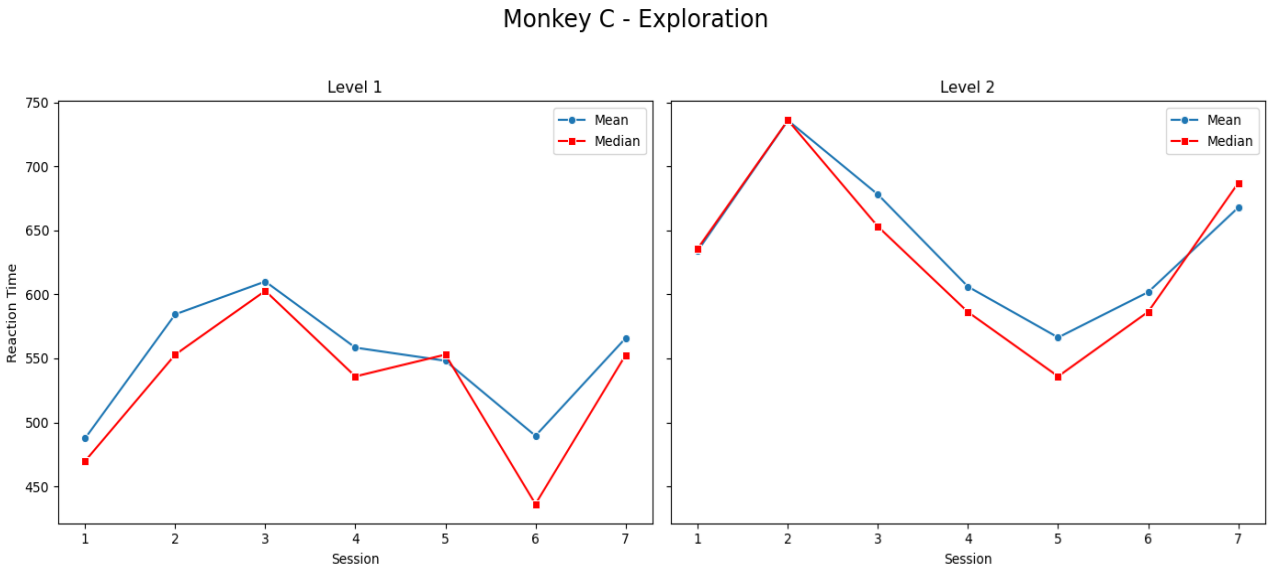

Figure 7. *Monkey C. Reaction time for Exploration trials across session for both levels (mean in blue, median in red). RT at level 1 is more stable and consistent across session (6 blocks each) than at level 2.*

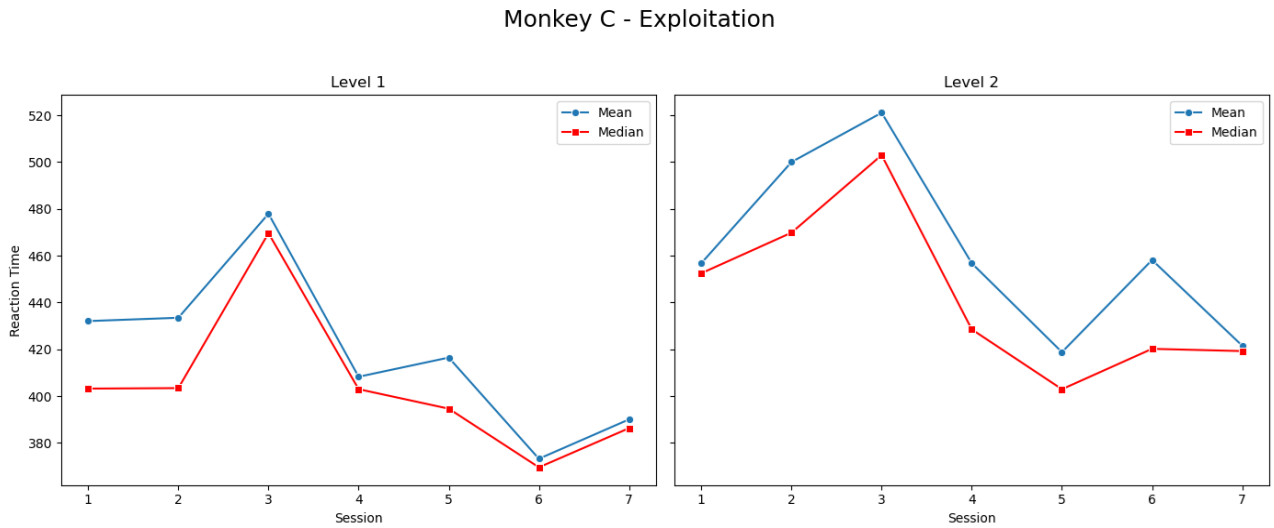

Figure 8. *Monkey C. Reaction time for Exploitation trials across session for both levels (level 1 in blue, level 2 in dashed orange).*

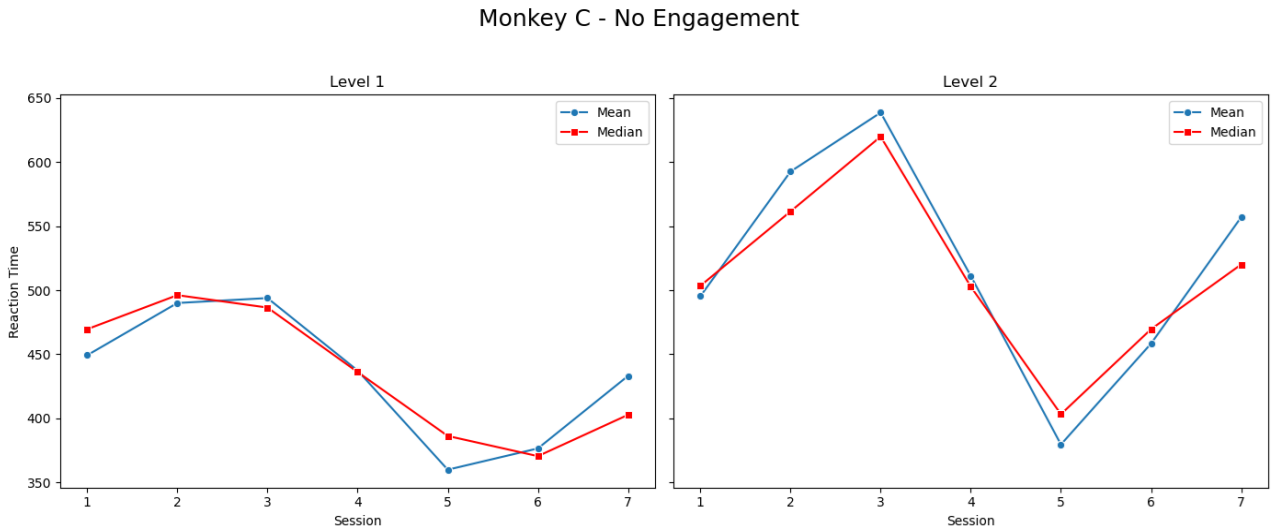

Figure 9. *Monkey C. Reaction time for No Engagement trials across session for both levels (level 1 in blue, level 2 in orange).*

4.2.3 Alternative Cognitive Efficiency: tracking and spatial strategies

The evidence that reaction times are shorter at Level 1 and longer at Level 2 may be driven by trial category. In fact, only once the monkey has made its first decision it can become aware of what category that trial represents. Depending on its Level 1 choice, it may require more or less time, especially if it needs to maximize reward because it's unsatisfied with the initial choice, or conversely, if it's already selected the most rewarding long-term option (the "low path"), it may disregard further deliberation.

An important observation is that reaction times faster to the side of the touch at Level 1 while fixation times lower. This difference is particularly shown by category.

Categories with low rewards (i.e., highlow and highhigh) both have nearly 80% of trials where fixations at the touched side are shorter at level 1 than level 2 (against the 60% for categories with high reward; lowhigh and lowlow). We run a repeated-measure ANOVA for each category individually and found that the difference between levels for categories highlow and highhigh was significant ($p < .001$).

In particular, the distinction between "in-series" and "not-in-series" helps us gauge how reaction time differs when the animal is certainly engaged ("in series") versus possibly disengaged. We therefore analyzed reaction times by category to see whether

differences emerged between levels and/or between in-series and not-in-series trials. The results (Figure 10) are especially intriguing: they reveal a clear Level 1 vs. Level 2 difference in the low-reward categories (high→low, high→high), but no marked difference between engagement states, almost as if the engagement level itself dictates which task level the decision is made at.

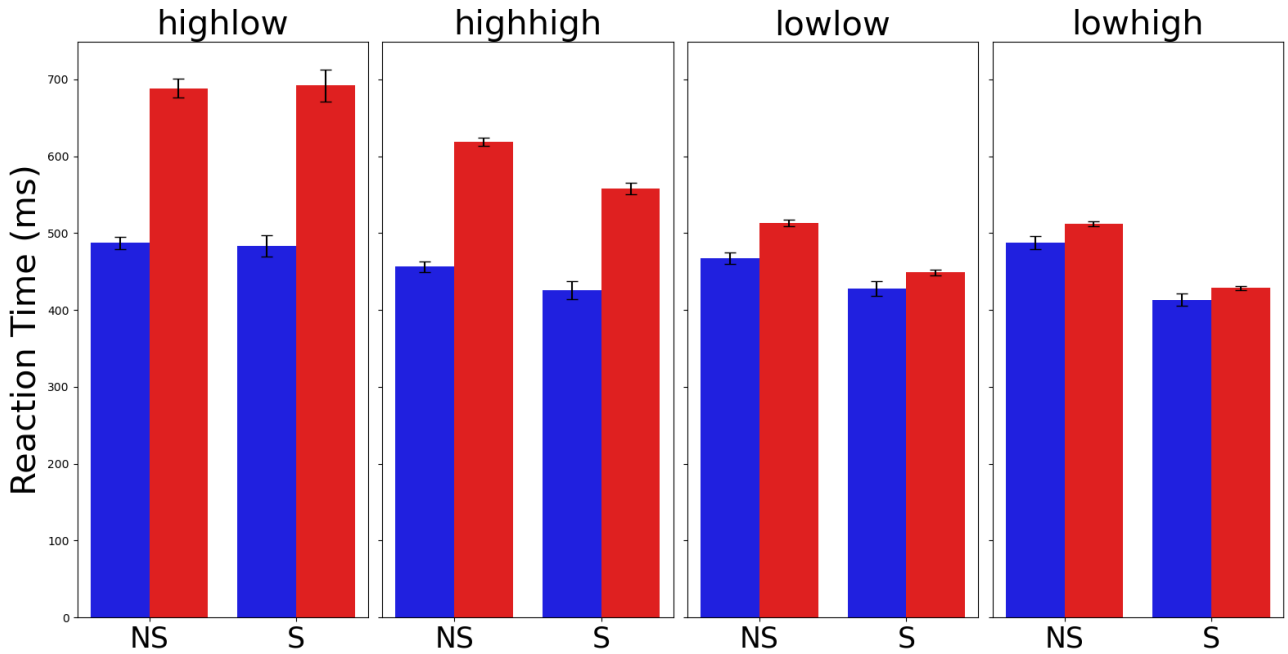

Figure 10. *Monkey C. Reaction Time by level and condition (serie/no serie) for each category (highlow, highhigh, lowlow, lowhigh). From left to right the global reward for each trial increases (highlow gives the lowest reward, lowhigh the highest).*

This pattern becomes even more interesting when compared across categories: as the potential reward increases, the gap between Level 1 and Level 2 shrinks. Although Level 1 reaction times remain consistent across categories, at Level 2 they decrease and no longer differ significantly from Level 1. Most notably, as reward increases, Level 2 reaction times drop, while the difference between in-series and not-in-series trials grows. In not-in-series conditions, reaction times are slower than in-series, a contrast that is not apparent in the lower-reward categories.

We therefore asked whether, by looking at the total reward per block, the gap between engagement and no-engagement was really so large. In other words, do trials labeled “no engagement”, a label driven by random choices rather than by maintaining a

strategy, truly reflect a state of animal laziness? Is sticking to a strategy always better than making random choices?

Indeed, Figure 11 shows that in some blocks, choosing at random, or using a "cognitive-saving" strategy (e.g. always selecting the same stimulus at Level 1 and randomizing at Level 2, or vice versa), yielded substantially higher rewards than persisting in a sub-optimal fixed sequence. In Figure 11, blocks are ordered by a descending reward rate. Notice that blocks with few in-series trials often rank among the highest-reward blocks in a session (in some cases they are the top block—see Sessions 4 and 6). For example, Block 3 of Session 1 and Block 1 of Session 2 show total rewards nearly matching the session's best block, despite very few "in-series" trials. The key difference, however, lies in the number of in-series trials (which indicate high engagement).

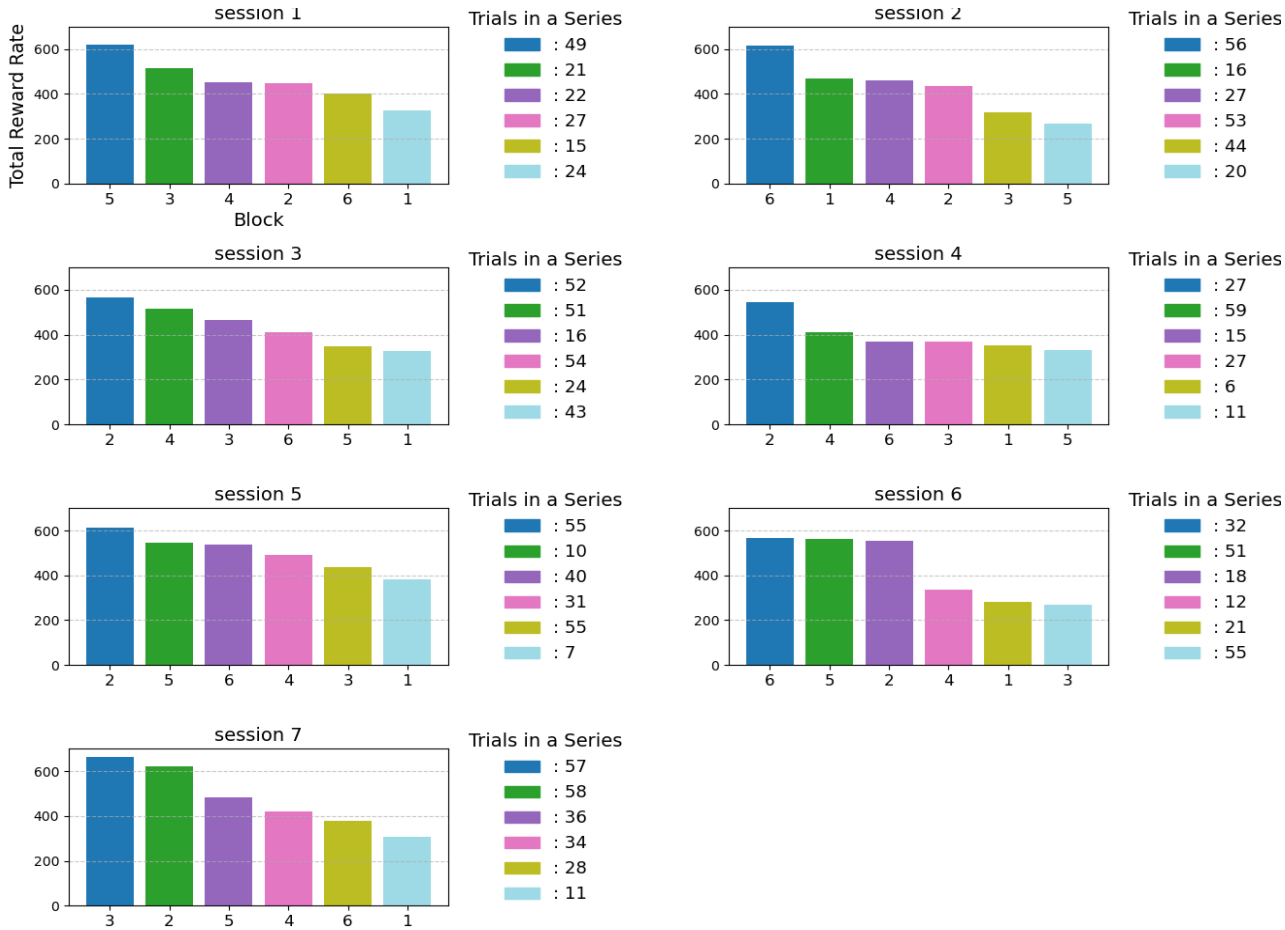

Figure 11. *Monkey C. Reward rate for each block in each session, looking at how many trial (out of 60) were "in series.*

Even more strikingly, some blocks with many in-series trials produced very low rewards (e.g. Block 3 of Sessions 2, 5, and 6). Conversely, as we initially hypothesized, blocks with very few in-series trials often led to very poor rewards, but at the same time, other highly rewarding strategies could be executed without any "in-series" choices. One such strategy is "level-saving," where the monkey always chooses the same stimulus in one level but randomizes in the other, saving cognitive effort on only one stimulus at two levels.

Thus we asked: can we distinguish true disengagement from a deliberate, random-choice strategy? First, we must define "true disengagement." To do so, we calculated how much reward an agent would accumulate per block if it made only random choices:

$$\text{Reward per block}=60\times[0.5\times(vt1+0.5vt3|t1+0.5vt4|t1)+0.5\times(vt2+0.5vt3|t2+0.5vt4|t2)]$$

where v denotes value, t1, t2 the Level 1 stimuli and t3, t4 the Level 2 stimuli. Based on the reward associated with each complete choice (both levels), a purely random-choice block would yield on average 344.5 drops (SD ± 20). That's far below the mean reward per block observed for blocks with fewer than half of their trials "in series" (< 30), which is 415.4 drops (SD ± 60). This suggests the monkey's "random" decisions are not purely random but reflect a deliberate strategy.

This average reward increases markedly for blocks with more than half their trials "in series" (> 30), to 476.6 drops (SD ± 91), but with considerable variance. What this highlights is that the animal consistently seeks a low cognitive-cost path to high reward: its overall reward is inevitably lower than if it had stuck with the single most rewarding sequence, but with the long-term expectation of saving cognitive resources.

Another important point to consider is the presence in Monkey C's behavior, of a spatial strategy which means choosing always the stimulus on the left or on the right, at one or both levels. As can be depicted in Figure 12, showing a heatmap of the percentage of touch on one side whether it was at level 1 or at level 2, in 10 blocks (out of 42) more than 70% of the choices fell systematically on one side. In almost 60% of the total amount of blocks, more than 60% of choice was done touching continuously on one side.

Overall, it is possible to highlight two major different strategies implemented: a tracking one, where the animal seeks a specific reward, and spatial one, where the animal does not care or partially care (in the case of one-level-tracking) about the reward, but picks always the choice presented at a preferred side.

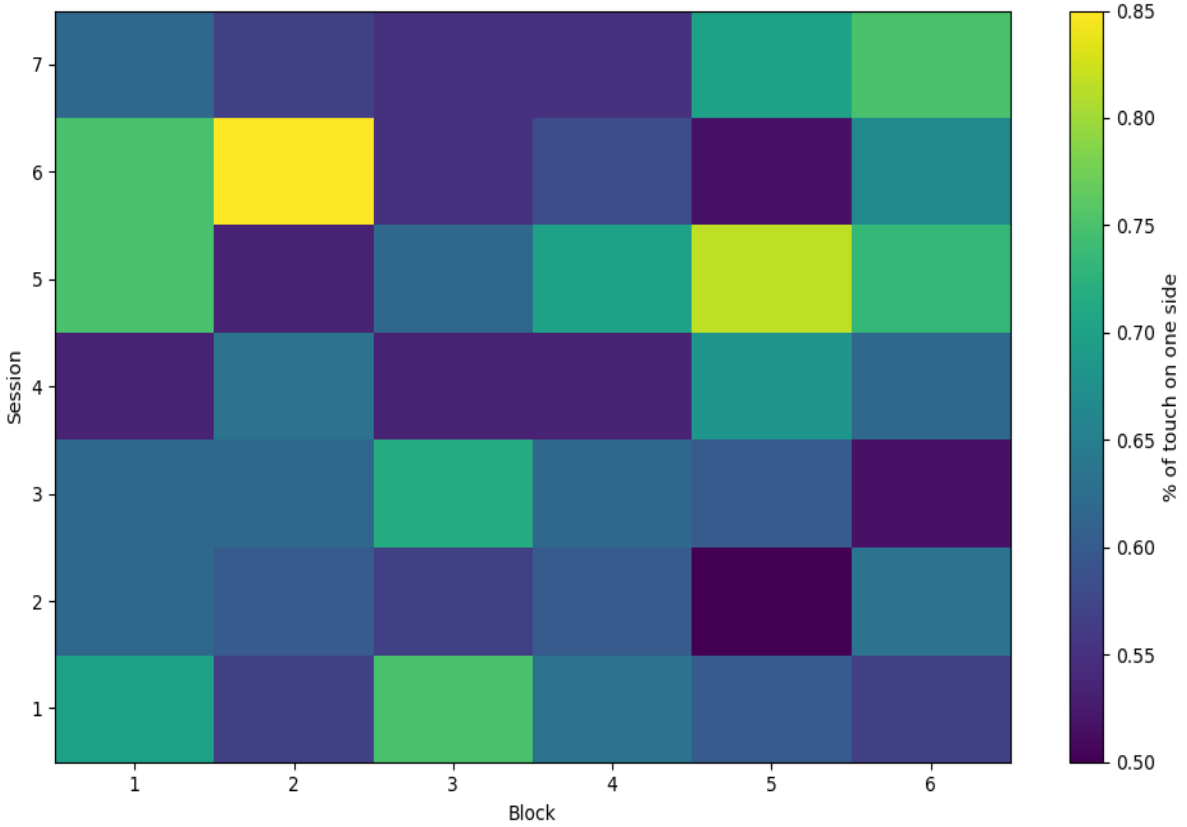

Figure 12. *Heatmap of the percentage of touch on one side at either level 1 or level 2. On the y-axis the sessions, on the x-axis, the blocks. Lighter blocks exhibit higher percentages of side biases.*

4.3 Discussion

The engagement state of an animal during decision-making tasks is fundamental to understanding the strategies it adopts and how it switches between them. We employed a two-level reverse-reward contingency task, known as the *consequential* task, to analyze whether and how macaques learn the task structure and use it to maximize rewards. In interpreting the monkeys' behavior, one useful perspective is the classic exploration–exploitation trade-off. For example, a subject may "exploit" known high-value stimuli while the other "explores" alternative targets more often, which matches decision-making models of explore/exploit dynamics. Daw and colleagues illustrated this dilemma in a slot-machine task: choosing the richest known option versus trying less familiar options to gain information and improve future rewards. Such exploration is not mere curiosity but is often critical for discovering the best strategies to harvest rewards (160). By observing behavior during the task, we extracted quantitative correlates of engagement states: exploration, exploitation, and disengagement.

4.3.1 Task Design and Implications on Engagement

At the core of the task was a manipulation of reward contingency: choosing the lower reward (1 drop) at the first level granted access to higher rewards at the second level (up to 9 drops), while choosing the higher initial reward (3 drops) led to minimal or no rewards at the second level. This setup encouraged the ability to delay gratification. The main methodological divergence between Monkey B and Monkey C concerned the criteria for block transitions. For Monkey C, blocks changed after 60 correct responses, regardless of the adopted strategy. For Monkey B, however, the transition occurred only after 30 consecutive choices within the same category.

4.3.2 Methodological Disparities and Cognitive Consequences

Monkey C, experiencing regular block changes every 60 correct responses, displayed a clear behavioral differentiation between exploration, exploitation, and disengagement. The predictable and marked changes in stimuli promoted the development of strategies, allowing the animal to maintain high levels of participation and strategic flexibility.

In contrast, Monkey B, subject to transitions dependent on rigid repetition of a choice, did not receive external signals prompting re-evaluation of its decisions. This led to poor task engagement, with an error rate of 72.57%, indicating low involvement or a lack of understanding of task dynamics. Monkey C, on the other hand, exhibited only a 2.76% error rate, confirming a solid grasp of the task structure and sustained engagement. Although this task difference may seem minor, as we have shown, it has a considerable effect on the animal's level of engagement. Furthermore, Monkey B completed two blocks in only 6 out of 12 sessions; in 3 sessions it completed one block, and in 3 sessions none at all. At the really beginning blocks were introduced to force the monkey to change strategies: since the animal kept doing only the same thing repetitively, the introduction of fixed changes in the stimuli was the solution to make him switch among categories in a flexible way. Thus, the task structure likely failed to keep the animal sufficiently engaged to complete multiple blocks, or else the animal did not grasp the mechanism by which stimuli changed (unlike in Monkey C's task, where one can learn that after a certain number of decisions, i.e., 60, the stimuli will change). Consequently, it stops working, a phenomenon that may resemble decision avoidance, in which the individual fully accepts its situation without interest in understanding its environment (161). More generally, decision avoidance strategies allow an animal to bypass difficult choices to reduce effort or regret. Recent human studies define decision-avoidance as postponing or delegating effortful decisions, which can reduce anticipated regret by evading responsibility for bad outcomes (162). This may explain why, in 3 sessions, even after finishing one block (albeit after many trials; mean = 114.3), the monkey ceased to work. On the other hand, this behavior could reflect perseveration and the associated switch cost: the monkey may habitually persist with previously learned response patterns and find it cognitively taxing to switch strategies when conditions change, leading it to give up rather than adapt to new contingencies. This is in line with studies (163) where cost switches act as a penalty that makes repeating the same strategy more efficient rather than changing it. This discrepancy can be interpreted as a difference between external and internal cognitive control. Monkey C's task structure, structured with fixed block changes, stimulated engagement through external feedback. Monkey B's task, by contrast, required intrinsic motivation, which is harder to maintain consistently throughout the task.
The design of our two-level task likely amplifies cognitive control demands. Multistep or

rule-switching tasks require top-down coordination, and studies of task switching show that performing frequent switches activates a control network (lateral prefrontal and parietal cortices) and incurs “switch costs” from lingering representations of the previous task (163). These data support hierarchical-control models in which some prefrontal areas maintain task rules while others resolve conflicts between competing actions. In our case, maintaining multiple levels of contingency or reversing reward mappings probably increased conflict, so that cognitive control had to offset interference. Theoretical accounts like Shenhav et al.’s Expected Value of Control (EVC) model predict that subjects will invest control only if the expected payoff outweighs the effort cost (165). Thus, the differences we saw may reflect how each monkey evaluated the cost–benefit of exerting sustained control under our task conditions.

4.3.3 Engagement States: Behavioral Metrics as a Window into Cognitive Strategy

To infer internal states from observable behavior, four main variables were measured: reaction time, fixation duration, number of saccades, and movement vigor. In Monkey C, during exploration phases, longer reaction and fixation times were recorded, as well as a greater number of saccades compared to exploitation. This supports the idea that exploration requires active information gathering, while exploitation reflects quick, confident decisions. Performance also depends on momentary cognitive engagement. In particular, shifts into an “off-task” or mind-wandering state would slow responses and increase errors. Mind wandering is characterized by attention switching away from the current task toward unrelated thoughts. This state involves “perceptual decoupling” and reduced executive control, leading to performance costs (e.g. greater errors) even though it can have long-term benefits like creativity or future planning (166).
No significant variation was observed in movement vigor. However, we observed a gradual decline in vigor over the course of each session, indicating that vigor tracks long-term changes in the animal’s state rather than trial-by-trial cognitive strategy. Moreover, comparisons between exploration and disengagement showed significant differences, while differences between exploitation and disengagement were absent, raising questions about how disengagement should be labeled or, more deeply, how

disengagement should be considered, which could eventually represent, as later theorized, strategic indifference.

In Monkey B, such distinctions were absent. This indicates that, if the structure of the task is not fixed enough, behavioral metrics cannot reliably serve as proxies for cognitive state.

Monkey B's behavior reflects this pattern: it does not explore different categories but instead tends to stick with whichever option happened to be most rewarding early on, considering both levels. From a reward-maximization and effort-minimization perspective, an optimal strategy would be to remain in a semi-engaged state, alternating between low and high reward at the first level and sticking with the same reward (e.g., high) at the second, regardless of category and ignore errors that might signal a higher payoff, provided the task remains linear and free of abrupt changes in stimuli or reward contingencies.

In Monkey C's task, a block switched as soon as the monkey accumulated 60 correct responses—no matter which category they came from. Although this rule should have prevented any category-based switching strategy, Monkey C is able to apply it on a smaller scale: he often persists with one category (or alternates between them) for fewer trials and successfully triggers block changes this way.

4.3.4 Behavioral Changes Between Levels: Temporal Dynamics and Decisions

An interesting phenomenon emerged: Monkey C showed faster reactions and shorter fixation time to the side of the touch at the first level compared to the second, even though the first level determined the reward trajectory. In particular, the huge difference between levels is shown depending on the category. Categories with low rewards (i.e., highlow and highhigh) both have nearly 80% of trials where fixations at the touched side are shorter at level 1 than level 2 (against the 60% for categories with high reward; lowhigh and lowlow). We run a repeated-measure ANOVA for each category individually and found that the difference between levels for categories highlow and highhigh was significant ($p < .001$). This suggests that, once the winning strategy was internalized, the animal acted decisively and rapidly, while the second level involved more reflective evaluation. These patterns are consistent with hierarchical reinforcement learning

mechanisms, where early decisions can become automatized while later stages require continued goal-directed deliberation (167, 168).

Supporting this, prior work in non-human primates has shown significantly faster responses at the first stage than the second in two-level choice tasks, especially when high rewards were anticipated, indicating a strategic prioritization of reaching the second level (23). Similarly, humans and animals adjust their speed–accuracy trade-off to optimize reward rate, acting faster when high-value outcomes are expected and slowing down when decisions require greater caution (169, 170). Thus, faster first-level responses in our task may reflect the monkey's understanding that the major reward was delivered at the second level.
Following choices that led to low rewards at the second level, reaction times increased, possibly due to risk of receiving no reward following the touch of high reward at the first level (and low at the second). Conversely, when the first-level choice had led to a "correct" path (i.e., the lower reward unlocking the high-reward trajectory), the second level was approached quickly and confidently. This may reflect a learned model-based strategy in which the subject predicts future outcomes based on current structure (168, 169)

In these cases where the reward was high at the first level, higher reaction times at level 2 may reflect a cautious behavior in order not to take no reward at level 2. This is consistent with the findings that subjects increase decision thresholds (i.e., become slower but more accurate) following low-value or error feedback, in order to maximize subsequent outcomes (171). The behavior may suggest a specific involvement at level 2 so that the animal quickly picked a choice at level 1 in order to get to the second level as fast as possible. This pattern can be seen as evidence that the animal understood the hierarchical structure of the task and that value-sensitive adjustments in response strategy were made to optimize reward (23, 169).
In Monkey B, these effects were absent, further confirming the lack of a flexible strategy and hierarchical task interpretation.

4.3.5 Strategic Variability and Reward Optimization

A surprising result concerns the relationship between behavioral strategy and reward rate: the most profitable phases did not always coincide with apparent engagement or

exploitation. In some cases, seemingly random or indifferent behaviors led to high rewards, suggesting the use of heuristics that simplify decision-making without maximizing each individual reward (165, 172). For example, when the animal is choosing the low target at level 1 and then alternate between high and low target at level 2. This may be due to the implementation of a side bias at level 2 and the randomized display of the target.

This raises questions about what truly defines cognitive engagement: repeated choices, exploratory effort, or reward maximization? Each criterion leads to different conclusions (173). An animal adopting different strategies but earning many rewards might appear disengaged, yet may be using an efficient cognitive rule (174).

This result can be for a certain degree associated to the tendency of Monkey C to shift from a tracking strategy (i.e., sticking with one category, thus tracking the stimulus/i) to a spatial one (i.e., choosing one side to touch, either left or right). Doing so, the animal can jump from an engaged to a disengaged state whenever he wants, since following a spatial or semi-spatial (if still tracking one level) strategy doesn't affect in a completely negative way the earnings of reward. This behavior may reflect a form of meta-decision-making, in which the cost of cognitive labor is weighed against its potential benefit (175), consistent with models of rational inattention in which engagement increases only when the expected gains from precision exceed cognitive costs (176).

4.3.6 Decision Avoidance and Motivational Decline

A salient observation in Monkey B is its tendency to avoid making decisions after completing a block. This could indicate a motivational decline, in which the effort required to maintain engagement exceeds the perceived value of the reward. In the absence of clear progress signals or feedback, the animal withdraws from the task, a behavior resembling learned helplessness.

Monkey C, instead, operated within a structured and reinforcing context. Predictable transitions, consistent rewards, and salient feedback encouraged sustained engagement. This highlights the importance of a well-designed environmental structure to support cognitive effort over time.

4.3.7 General Implications and Future Perspectives

This study demonstrates how small methodological differences, such as block transition rules, can have large impacts on cognitive engagement in non-human primates. It also shows how detailed behavioral metrics can be employed to infer internal states such as exploration, exploitation, or disengagement.
In conclusion, cognitive engagement emerges as the dynamic interaction between task design, motivation, and individual strategy. Behavioral metrics are valuable tools for probing these states but require a robust experimental context.

4.4 Limitations

Despite the results obtained, the study presents several limitations at different levels of task design and implementation.

A first limitation concerns the comparability of analysis across two substantially different tasks: Monkey C performed a task in which block changes occurred every 60 trials, while Monkey B experienced the same change only after 30 consecutive trials choosing the same category. As discussed in the results and discussion sections, this task changes greatly influenced the engagement and strategies adopted by the two monkeys. Therefore, a mirrored comparison of strategies between the two animals should be made using identical tasks to avoid effects stemming from the structural differences in the task. The difference in the monkeys' behavior also impacted on the analyses. The labeling of engagement states was thus inconsistent between the two animals, as the task differences led to opposite behaviors: on one side, sustained engagement across many trials (Monkey C), and on the other, strong initial engagement that was quickly suppressed (Monkey B).

Another limitation lies in the labeling of engagement states. As highlighted in the methods, labeling is arbitrary, and classification conditions were defined based on behaviors and strategies observed during task execution. Moreover, as emerged from the results, behavioral variables only explain some of the differences between engagement states, raising doubts about the classification. As we explained, what may appear externally as disengagement might instead reflect a latent strategy aiming to maximize gain despite low motivation. Similarly, exploration could in some blocks be absent (with total disengagement and disinterest in the task), more limited than classified, or more extensive, depending on how many trials the animal needs to learn the stimulus-reward

association. Indeed, we do not exclude that the animal's learning rate could vary from block to block, not due to the difficulty of stimuli (since there was no direct classification of stimuli as complex/simple), but due to the animal's intrinsic motivation and thus its engagement with the task. Finally, the behavioral variables acquired during task recording may not be sufficient to determine engagement states. Pupil dilation, for example, was not recorded but could have been important for categorizing animal behavior.

Future studies should focus on a more methodical classification of engagement and assess tasks across more than one animal to understand whether individual differences exist in the adoption of heuristics or whether the pattern remains consistent. In particular, integrating neural data could allow for physiological validation of engagement states.

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